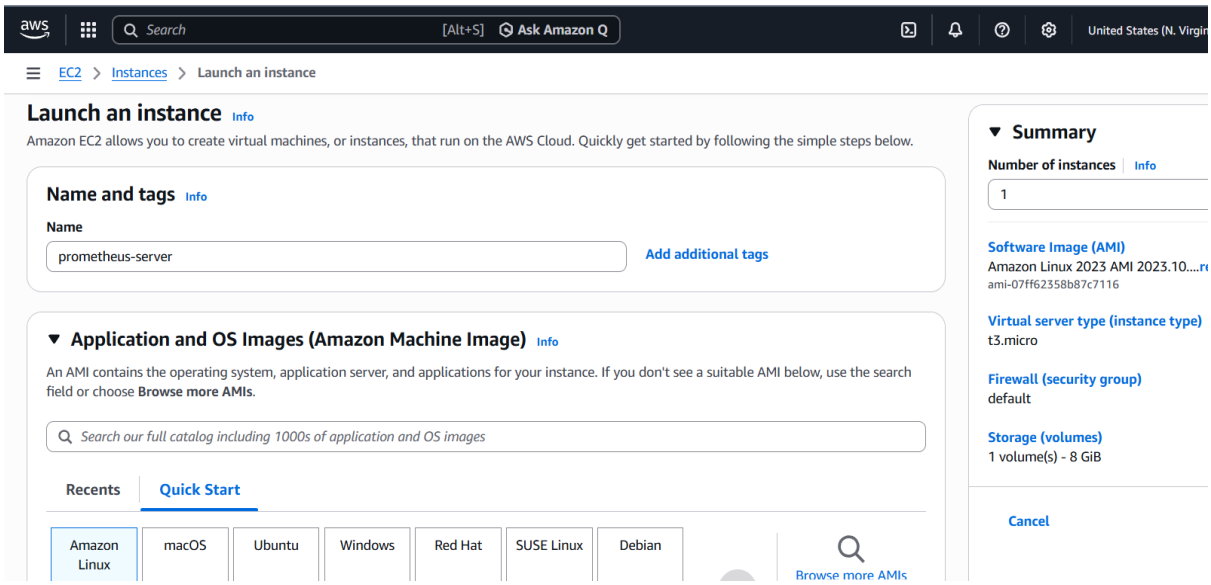
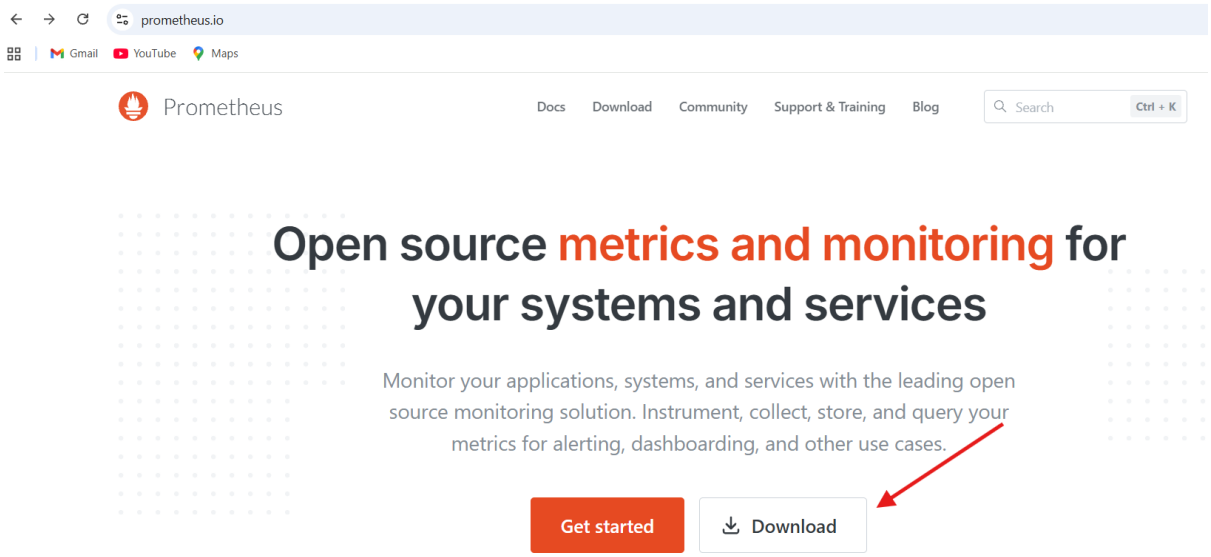
1. Install Prometheus on a server and configure prometheus.yml.

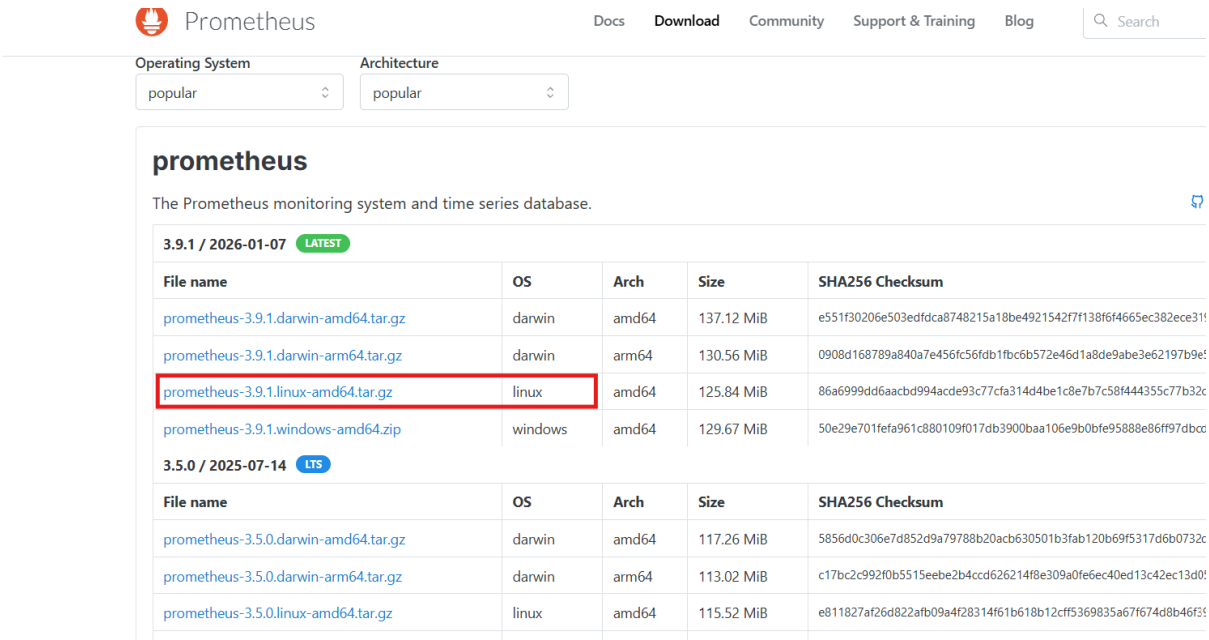
Launch a ec2-instance



Go to Prometheus.io site



Copy this linux link



Install in the server and extract the file.

```
[root@ip-172-31-88-247 opt]# wget https://github.com/prometheus/prometheus/releases/download/v3.9.1/prometheus-3.9
--2026-01-21 07:08:33--  https://github.com/prometheus/prometheus/releases/download/v3.9.1/prometheus-3.9.1.linux-
Resolving github.com (github.com)... 140.82.114.4
Connecting to github.com (github.com)|140.82.114.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/66ca4596-6d94-4fe8-
11-09&sr=b&spr=https&se=2026-01-21T07%3A43%3A16Z&rscd=attachment%3B+filename%3Dprometheus-3.9.1.linux-amd64.tar.gz
m&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-21T06%3A42%3A4
&sks=b&skv=2018-11-09&sig=U01hWeoPgK6gz%2FjOnHTu8fZoRdYhjrD%2B8oh7KxpDEVU%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1N
iYXVkIjoicmVsZWFzZS1hc3NldHMuZ2lOaHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc2ODk4Mjg2OSwibmJmIjoxNzY4O
c2VOcHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.sqEmeUCifBLZteM-B7ZO9h5hGCkAmw-A9wGij9nDCVU&response-content-di
ame%3Dprometheus-3.9.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-01-21 07:08:33--  https://release-assets.githubusercontent.com/github-production-release-asset/6838921/66ca
1?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-01-21T07%3A43%3A16Z&rscd=attachment%3B+filename%3Dprometheus-3.9.1.lin
n%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01
1T07%3A43%3A16Z&sks=b&skv=2018-11-09&sig=U01hWeoPgK6gz%2FjOnHTu8fZoRdYhjrD%2B8oh7KxpDEVU%3D&jwt=eyJ0eXAiOiJKV1QiLC
aXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2lOaHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc2ODk4Mjg2OS
joicmVsZWFzZWFzZc2VOcHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.sqEmeUCifBLZteM-B7ZO9h5hGCkAmw-A9wGij9nDCVU&resp
ment%3B%20filename%3Dprometheus-3.9.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.110.133|:443...
HTTP request sent, awaiting response... 200 OK
Length: 131949217 (126M) [application/octet-stream]
Saving to: 'prometheus-3.9.1.linux-amd64.tar.gz'

prometheus-3.9.1.linux-amd64.tar.gz  100%[=====================================================================>] 12

2026-01-21 07:08:33 (444 MB/s) - 'prometheus-3.9.1.linux-amd64.tar.gz' saved [131949217/131949217]

[root@ip-172-31-88-247 opt]# ls
aws  prometheus-3.9.1.linux-amd64.tar.gz
[root@ip-172-31-88-247 opt]# tar xvf prometheus-3.9.1.linux-amd64.tar.gz
prometheus-3.9.1.linux-amd64/
prometheus-3.9.1.linux-amd64/NOTICE
prometheus-3.9.1.linux-amd64/LICENSE
prometheus-3.9.1.linux-amd64/prometheus.yml
prometheus-3.9.1.linux-amd64/promtool
prometheus-3.9.1.linux-amd64/prometheus
[root@ip-172-31-88-247 opt]#
```

Go to extracted file we are having Prometheus.yml file.

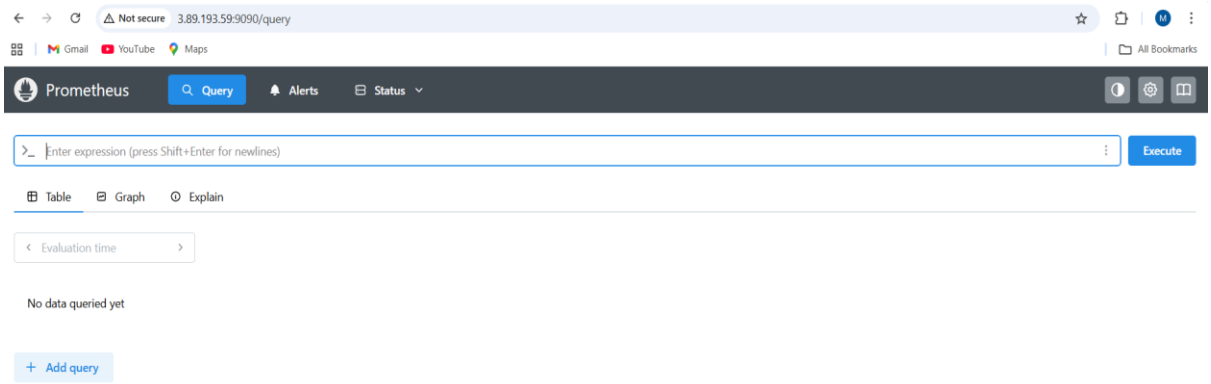
```
[root@ip-172-31-88-247 opt]# ls
aws  prometheus-3.9.1.linux-amd64  prometheus-3.9.1.linux-amd64.tar.gz
[root@ip-172-31-88-247 opt]# cd prometheus-3.9.1.linux-amd64/
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ls
LICENSE  NOTICE  prometheus  prometheus.yml  promtool
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]#
```

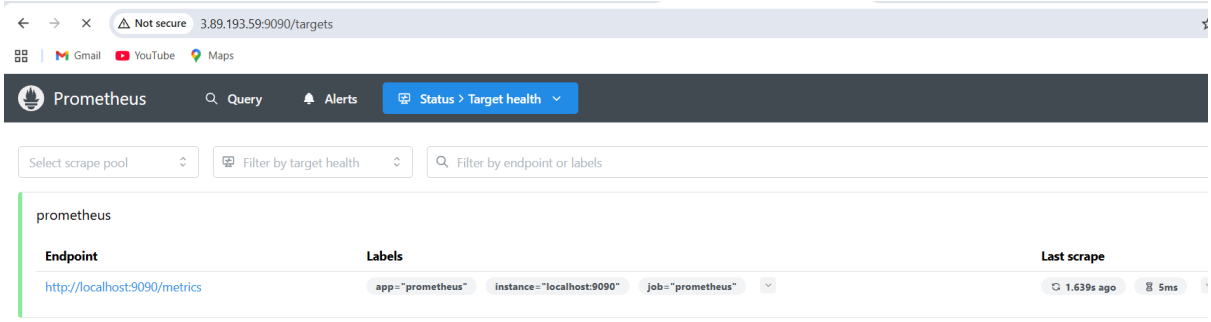
Execute this command to run the Prometheus server.

- **./prometheus --config.file=prometheus.yml**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ./prometheus --config.file=prometheus.yml
time=2026-01-21T07:14:00.214Z level=INFO source=main.go:1589 msg="updated GOGC" old=100 new=75
time=2026-01-21T07:14:00.222Z level=INFO source=main.go:704 msg="Leaving GOMAXPROCS=2: CPU quota undefined"
time=2026-01-21T07:14:00.223Z level=INFO source=memlimit.go:198 msg="GOMEMLIMIT is updated" component=autome
un/automemlimit/memlimit GOMEMLIMIT=865231257 previous=9223372036854775807
time=2026-01-21T07:14:00.223Z level=INFO source=main.go:752 msg="No time or size retention was set so using
15d
time=2026-01-21T07:14:00.223Z level=INFO source=main.go:803 msg="Starting Prometheus Server" mode=server ver
vision=9ec59baffb547e24f1468a53eb82901e58feabd8)"
time=2026-01-21T07:14:00.223Z level=INFO source=main.go:808 msg="operational information" build_context="(go
=root@61c3a9212c9e, date=20260107-16:08:09, tags=netgo,builtinassets)" host_details="(Linux 6.1.159-181.297.
C Mon Dec 22 22:31:59 UTC 2025 x86_64 ip-172-31-88-247.ec2.internal (none))" fd_limits="(soft=65535, hard=65
d=unlimited)"
time=2026-01-21T07:14:00.226Z level=INFO source=web.go:684 msg="Start listening for connections" component=w
time=2026-01-21T07:14:00.229Z level=INFO source=main.go:1331 msg="Starting TSDB ..."
time=2026-01-21T07:14:00.233Z level=INFO source=tls_config.go:354 msg="Listening on" component=web address=[
time=2026-01-21T07:14:00.233Z level=INFO source=tls_config.go:357 msg="TLS is disabled." component=web http2
time=2026-01-21T07:14:00.248Z level=INFO source=head.go:681 msg="Replaying on-disk memory mappable chunks if
time=2026-01-21T07:14:00.248Z level=INFO source=head.go:767 msg="On-disk memory mappable chunks replay compl
time=2026-01-21T07:14:00.248Z level=INFO source=head.go:775 msg="Replaying WAL, this may take a while" compo
time=2026-01-21T07:14:00.249Z level=INFO source=head.go:848 msg="WAL segment loaded" component=tsdb segment=
time=2026-01-21T07:14:00.250Z level=INFO source=head.go:885 msg="WAL replay completed" component=tsdb checkp
ay_duration=1.129134ms wbl_replay_duration=472ns chunk_snapshot_load_duration=0s mmap_chunk_replay_duration=
2ms
time=2026-01-21T07:14:00.261Z level=INFO source=main.go:1352 msg="filesystem information" fs_type=XFS_SUPER_
time=2026-01-21T07:14:00.261Z level=INFO source=main.go:1355 msg="TSDB started"
time=2026-01-21T07:14:00.261Z level=INFO source=main.go:1542 msg="Loading configuration file" filename=prome
time=2026-01-21T07:14:00.265Z level=INFO source=main.go:1582 msg="Completed loading of configuration file" d
```

Search in browser with the ip and port number 9090





2. Install and configure Node Exporter for system monitoring.

To run the service in background create service file for Prometheus.

- **vi /etc/systemd/system/prometheus.service**

```
[Unit]
Description=Prometheus Monitoring Service
After=network.target

[Service]
Type=simple
ExecStart=/opt/prometheus-3.9.1.linux-amd64/prometheus \
  --config.file=/opt/prometheus-3.9.1.linux-amd64/prometheus.yml \
  --storage.tsdb.path=/opt/prometheus-3.9.1.linux-amd64/data
Restart=always

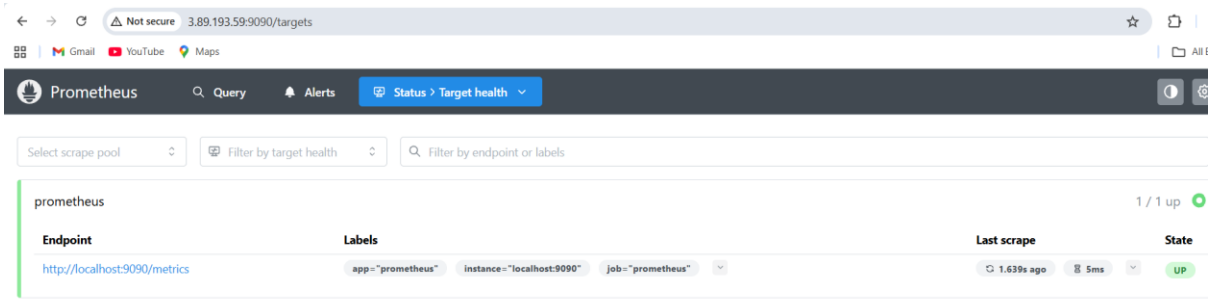
[Install]
WantedBy=multi-user.target
```

- **systemctl daemon-reload**
- **systemctl restart Prometheus**
- **systemctl status prometheus**

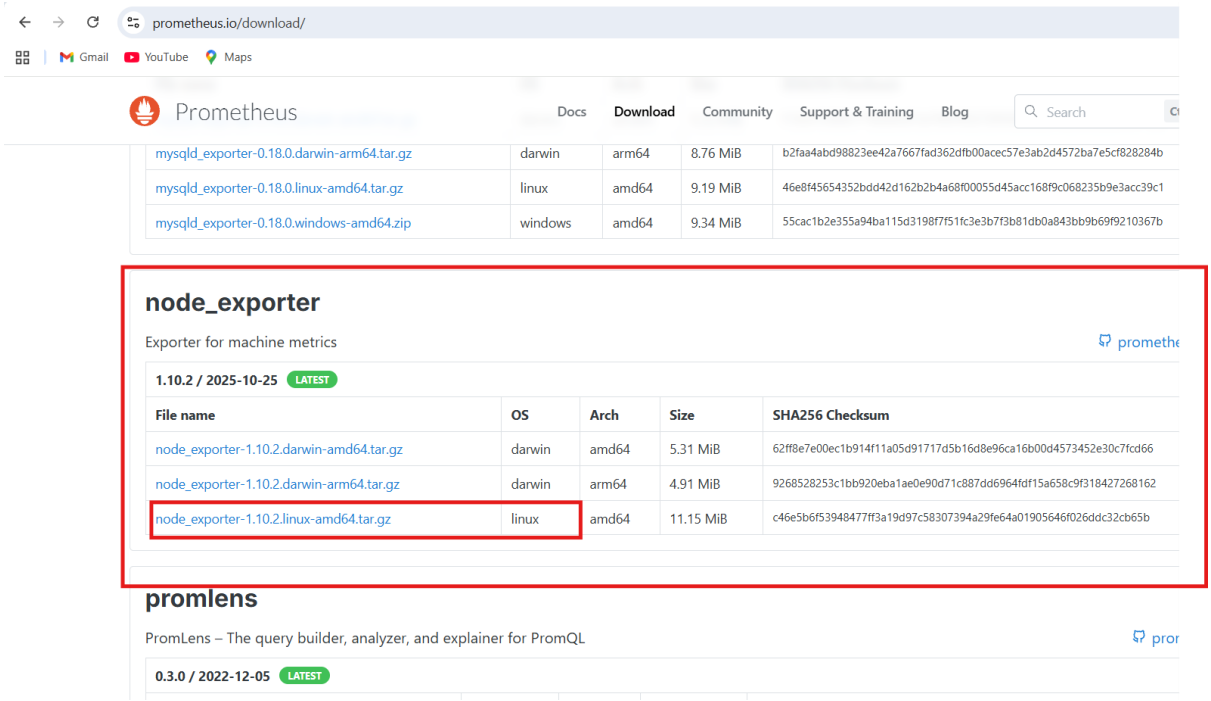
```
[root@ip-172-31-88-247 opt]# vi /etc/systemd/system/prometheus.service
[root@ip-172-31-88-247 opt]# systemctl daemon-reload
[root@ip-172-31-88-247 opt]# systemctl restart prometheus
[root@ip-172-31-88-247 opt]# systemctl status prometheus
● prometheus.service - Prometheus Monitoring Service
     Loaded: loaded (/etc/systemd/system/prometheus.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 07:45:33 UTC; 9s ago
   Main PID: 3303 (prometheus)
      Tasks: 8 (limit: 1067)
     Memory: 17.3M
        CPU: 92ms
     CGroup: /system.slice/prometheus.service
             └─3303 /opt/prometheus-3.9.1.linux-amd64/prometheus --config.file=/opt/prometheus-3

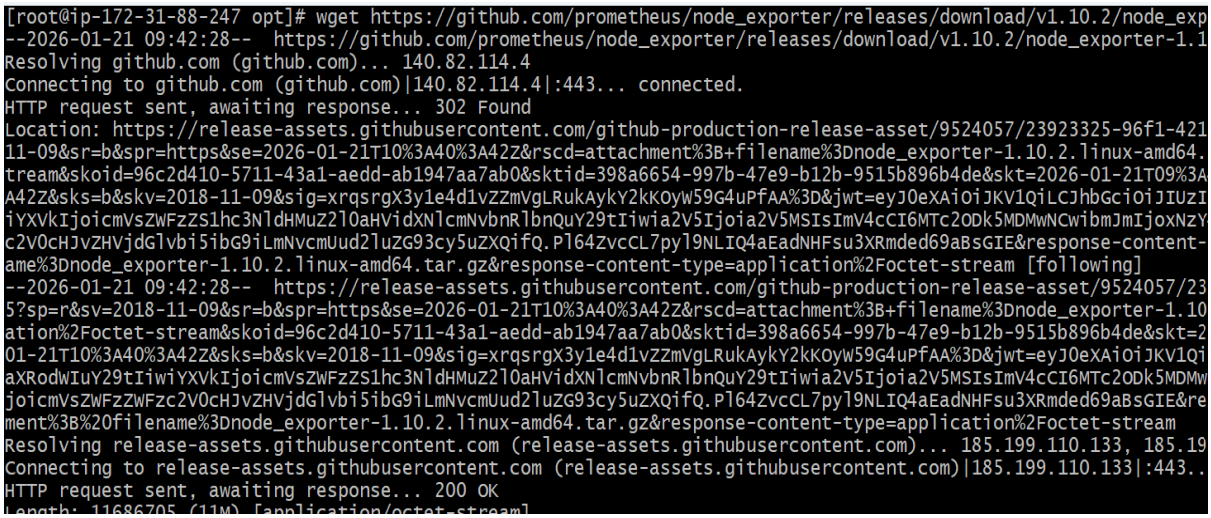
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.412Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.418Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.419Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.419Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.425Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.425Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.425Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.429Z le
Jan 21 07:45:33 ip-172-31-88-247.ec2.internal prometheus[3303]: time=2026-01-21T07:45:33.429Z le
```

It will not run only when the cursor is running it will run in the background .

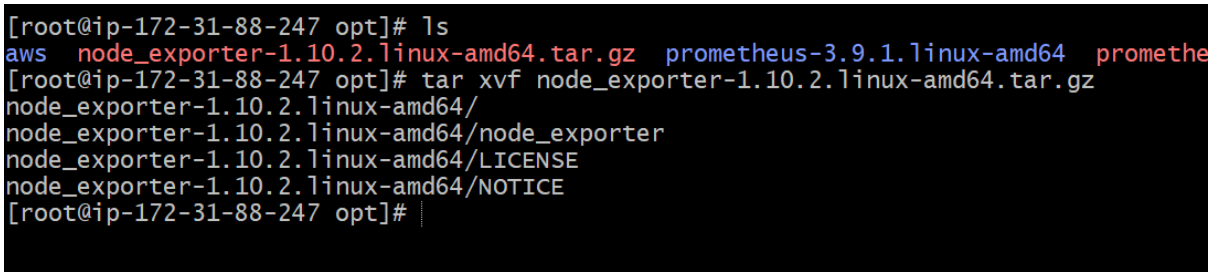


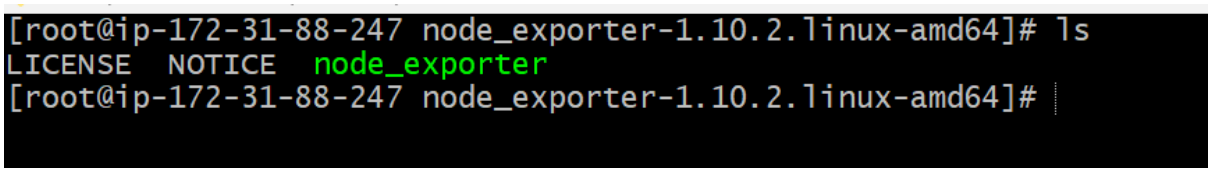
Copy the link of node exporter in official website and wget in the Prometheus server.





- tar xvf





inside the node exporter create a service file to run node exporter in background.

- **vi /etc/system/system/node_exporter.service**

```
[Unit]
Description=Node Exporter
After=network.target

[Service]
Type=simple
ExecStart=/opt/node_exporter-1.10.2.linux-amd64/node_exporter
Restart=always

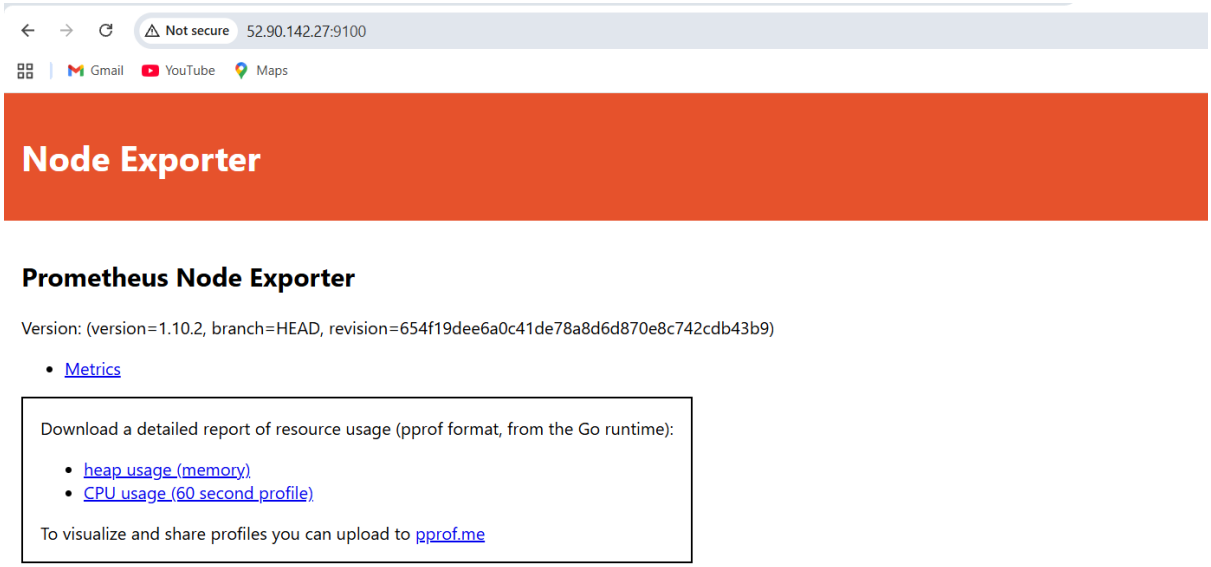
[Install]
WantedBy=multi-user.target
```

- **systemctl daemon-reload**
- **systemctl restart node_exporter**
- **systemctl status node_exporter**

```
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# vi /etc/systemd/system/node_exporter.service
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl daemon-reload
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl restart node_exporter
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl status node_exporter
● node_exporter.service - Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 10:01:28 UTC; 7s ago
   Main PID: 2880 (node_exporter)
      Tasks: 4 (limit: 1067)
     Memory: 2.6M
        CPU: 8ms
     CGroup: /system.slice/node_exporter.service
             └─2880 /opt/node_exporter-1.10.2.linux-amd64/node_exporter

Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
Jan 21 10:01:28 ip-172-31-88-247.ec2.internal node_exporter[2880]: time=2026-01-21T10:01:28.921Z level=INFO so
```

search in browser with the ip 9100



Connect node exporter to Prometheus

```
[root@ip-172-31-88-247 opt]# ls
aws                                node_exporter-1.10.2.linux-amd64.tar.gz  prometheus-3.9.1.linux-amd64.t
node_exporter-1.10.2.linux-amd64  prometheus-3.9.1.linux-amd64
[root@ip-172-31-88-247 opt]# cd prometheus-3.9.1.linux-amd64/
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ls
LICENSE  NOTICE  data  prometheus  prometheus.yml  promtool
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# vi prometheus.yml
```

- vi Prometheus.yml

change this file

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this config.
        labels:
          app: "prometheus"
```

Add this line near static configs

- **- tagets: [“localhost:9100”]**

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is ever
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

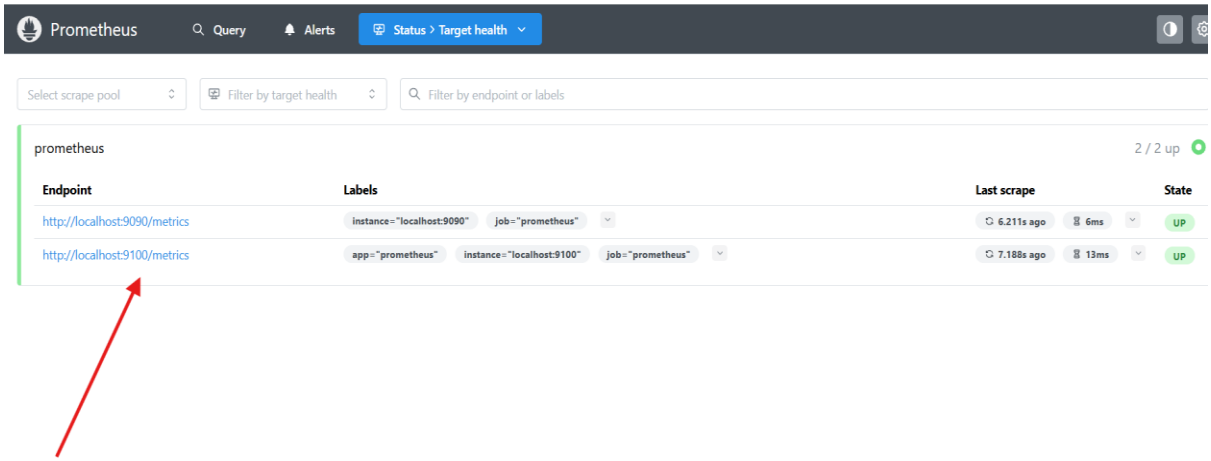
# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
       # The label name is added as a label `label_name=<label_value>` to any timeser
        labels:
          app: "prometheus"
~
~
~
~
~
```

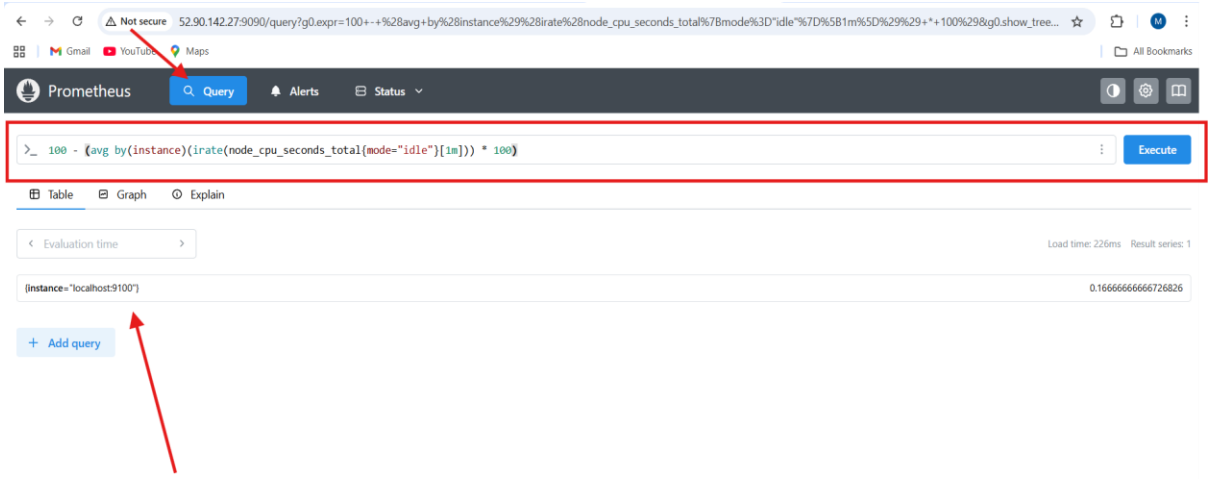
- **systemctl restart Prometheus**

go to dashboard and check the targets the node exporter is connected to Prometheus.



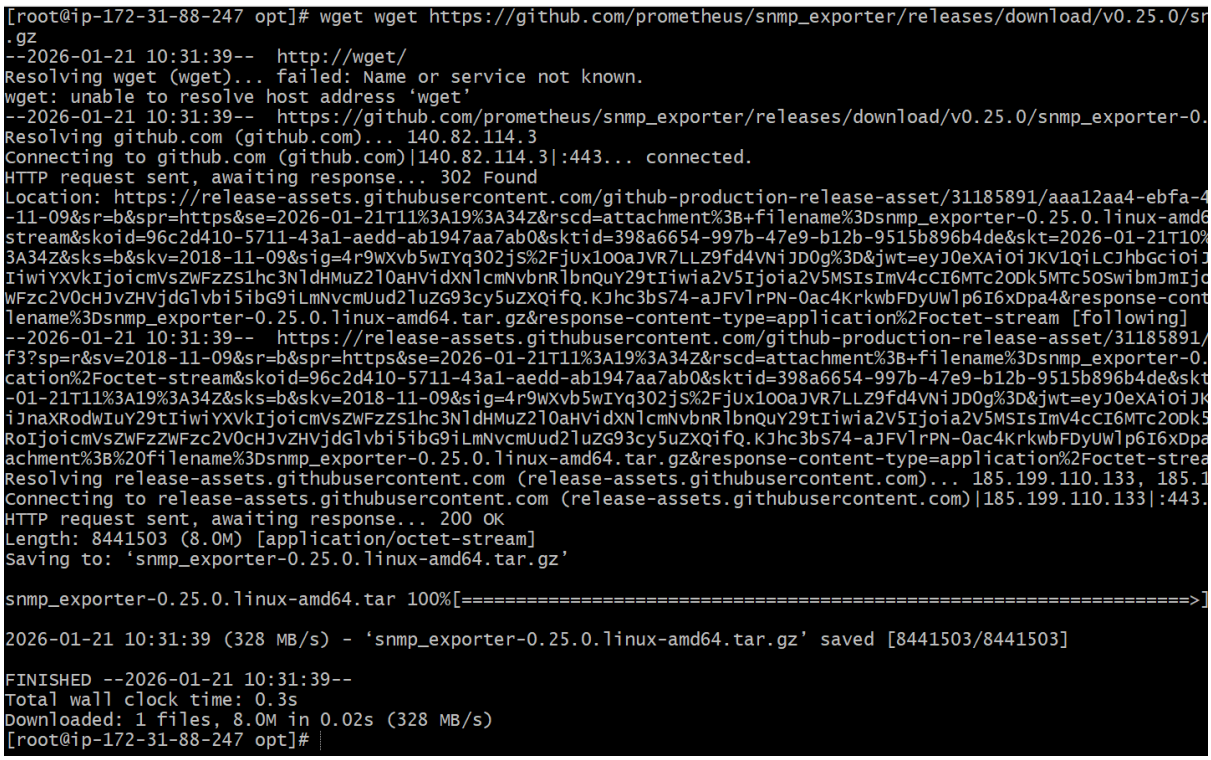
If you want to check the cpu utilization execute this command.

- **100 - (avg by(instance)(irate(node_cpu_seconds_total{mode="idle"}[1m])) * 100)**

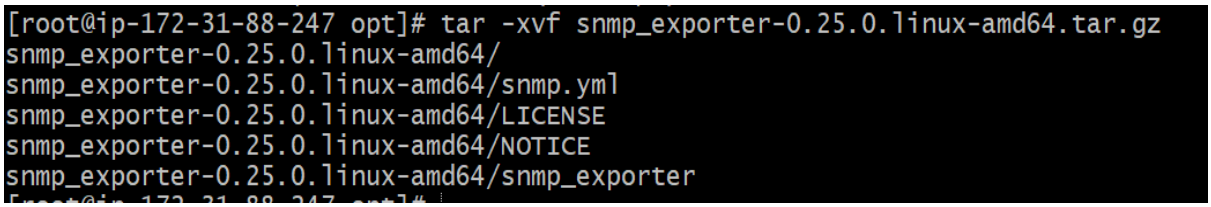


3. Set up SNMP Exporter to monitor network devices.

- **wget https://github.com/prometheus/snmp_exporter/releases/download/v0.25.0/snmp_exporter-0.25.0.linux-amd64.tar.gz**



- **tar -xvf snmp_exporter-0.25.0.linux-amd64.tar.gz**



Create a service file for snmp exporter

- **vi /etc/systemd/system/snmp_exporter**

```
[Unit]
Description=Prometheus SNMP Exporter
After=network.target

[Service]
Type=simple
ExecStart=/opt/snmp_exporter-0.25.0.linux-amd64/snmp_exporter \
  --config.file=/opt/snmp_exporter-0.25.0.linux-amd64/snmp.yml
Restart=always

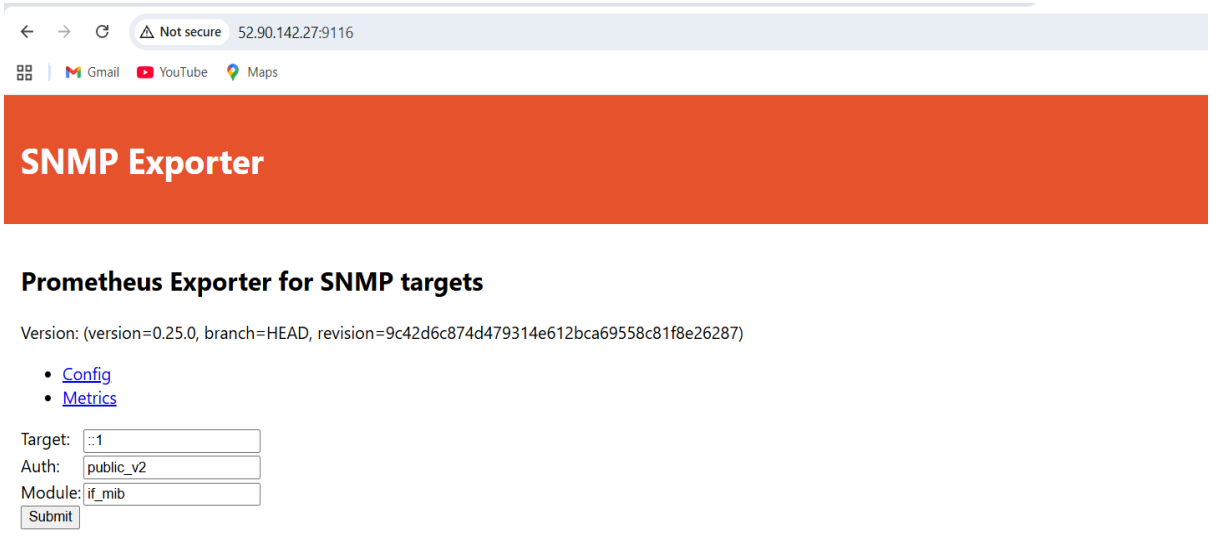
[Install]
WantedBy=multi-user.target
```

- **systemctl daemon-reload**
- **systemctl start snmp_exporter**
- **systemctl status snmp_exporter**

```
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# vi /etc/systemd/system/snmp_exporter.service
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl daemon-reload
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl start snmp_exporter
[root@ip-172-31-88-247 node_exporter-1.10.2.linux-amd64]# systemctl status snmp_exporter
● snmp_exporter.service - Prometheus SNMP Exporter
     Loaded: loaded (/etc/systemd/system/snmp_exporter.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 10:45:55 UTC; 4s ago
   Main PID: 4341 (snmp_exporter)
      Tasks: 7 (limit: 1067)
     Memory: 15.9M
        CPU: 93ms
     CGroup: /system.slice/snmp_exporter.service
             └─4341 /opt/snmp_exporter-0.25.0.linux-amd64/snmp_exporter --config.file=/opt/snmp_exporter-0.25.0.linux

Jan 21 10:45:55 ip-172-31-88-247.ec2.internal systemd[1]: Started snmp_exporter.service - Prometheus SNMP Exporter.
Jan 21 10:45:55 ip-172-31-88-247.ec2.internal snmp_exporter[4341]: ts=2026-01-21T10:45:55.388Z caller=main.go:194 lev
Jan 21 10:45:55 ip-172-31-88-247.ec2.internal snmp_exporter[4341]: ts=2026-01-21T10:45:55.388Z caller=main.go:195 lev
Jan 21 10:45:55 ip-172-31-88-247.ec2.internal snmp_exporter[4341]: ts=2026-01-21T10:45:55.464Z caller=tls_config.go:2
Jan 21 10:45:55 ip-172-31-88-247.ec2.internal snmp_exporter[4341]: ts=2026-01-21T10:45:55.464Z caller=tls_config.go:2
lines 1-15/15 (END)
```

Search on browser with the public ip and port number 9116.



Go to Prometheus server

- **vi Prometheus.yml**

Connect snmp_exporter to Prometheus

- **- targets: [“localhosts:9116”]**

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

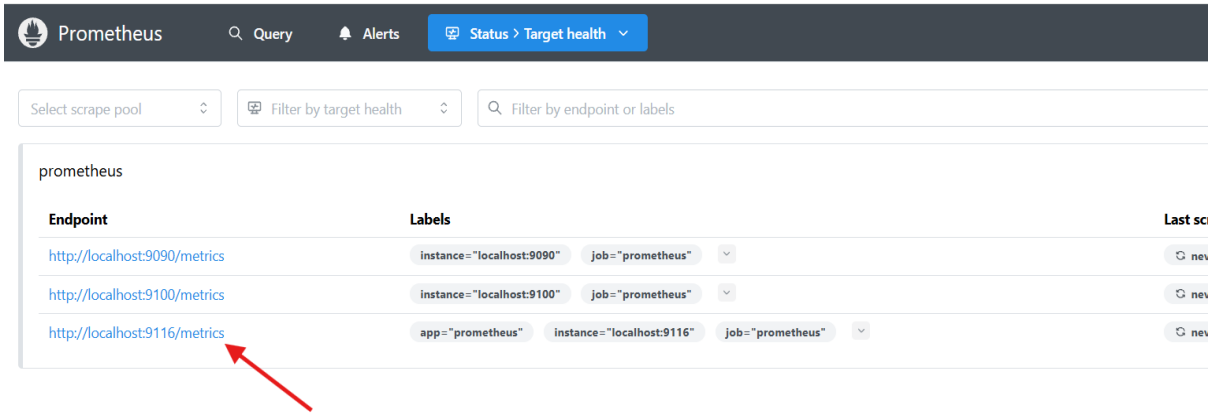
# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this
        labels:
          app: "prometheus"
~
~
~
```

- **systemctl restart Prometheus**

go to Prometheus server and go to dashboard and click on targets.



4. Set up Prometheus to run as a systemd service for automatic startup.

Create a service file Prometheus

- **vi /etc/systemd/system/Prometheus.service**

```
[Unit]
Description=Prometheus Monitoring
After=network.target

[Service]
Type=simple
ExecStart=/opt/prometheus-3.9.1.linux-amd64/prometheus \
  --config.file=/opt/prometheus-3.9.1.linux-amd64/prometheus.yml \
  --storage.tsdb.path=/opt/prometheus-3.9.1.linux-amd64/data
Restart=always

[Install]
WantedBy=multi-user.target
```

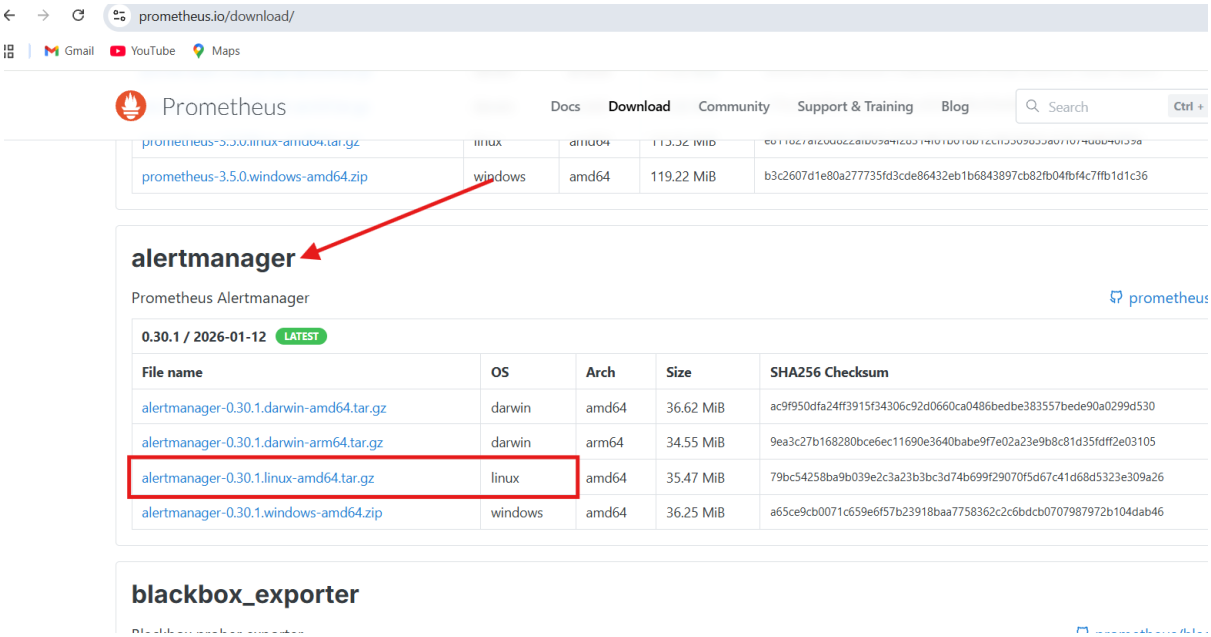
- **systemctl daemon-reload**
- **systemctl start Prometheus**
- **systemctl status prometheus**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ls
LICENSE  NOTICE  data  prometheus  prometheus.yml  promtool
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# vi /etc/systemd/system/prometheus.service
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl daemon-reload
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl start prometheus
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl status prometheus
● prometheus.service - Prometheus Monitoring
     Loaded: loaded (/etc/systemd/system/prometheus.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 10:57:30 UTC; 9min ago
   Main PID: 4686 (prometheus)
      Tasks: 8 (limit: 1067)
     Memory: 26.7M
        CPU: 702ms
     CGroup: /system.slice/prometheus.service
             └─4686 /opt/prometheus-3.9.1.linux-amd64/prometheus --config.file=/opt/prometheus-3.9

Jan 21 10:57:30 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T10:57:30.684Z leve
Jan 21 10:57:30 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T10:57:30.687Z leve
Jan 21 10:57:30 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T10:57:30.688Z leve
Jan 21 10:57:30 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T10:57:30.688Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.576Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.618Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.618Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.623Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.623Z leve
Jan 21 11:00:00 ip-172-31-88-247.ec2.internal prometheus[4686]: time=2026-01-21T11:00:00.631Z leve
lines 1-20/20 (END)
```

5. Install Alertmanager and configure an email alert.

Go to official website of Prometheus and copy link.




```
[root@ip-172-31-88-247 opt]# wget https://github.com/prometheus/alertmanager/releases/download/v0.30.1/al
--2026-01-21 12:07:03--  https://github.com/prometheus/alertmanager/releases/download/v0.30.1/alertmanage
Resolving github.com (github.com)... 140.82.112.4
Connecting to github.com (github.com)|140.82.112.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/11452538/ff9d26a3-
-11-09&sr=b&spr=https&se=2026-01-21T12%3A52%3A49Z&rscd=attachment%3B+filename%3Dalertmanager-0.30.1.linux
tream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-2
A49Z&sks=b&skv=2018-11-09&sig=JZkob%2B9Jlqh1yhaeRvokXK4zd6f492azdGaIzOSQY%2Bw%3D&jwt=eyJ0eXAiOiJKV1QiLCJh
tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2lOaHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc2ODk5OTAyMywi
ZWFzc2VOcHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.U2b3hNMq26cyzODOC4lOYjpY6kas61atigqeobLR3AI&respon
ilename%3Dalertmanager-0.30.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [follow
--2026-01-21 12:07:03--  https://release-assets.githubusercontent.com/github-production-release-asset/114
be?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-01-21T12%3A52%3A49Z&rscd=attachment%3B+filename%3Dalertmanag
ation%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4d
01-21T12%3A52%3A49Z&sks=b&skv=2018-11-09&sig=JZkob%2B9Jlqh1yhaeRvokXK4zd6f492azdGaIzOSQY%2Bw%3D&jwt=eyJ0e
OiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2lOaHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6M
XRoIjoicmVsZWFzZWFzZ2VOcHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.U2b3hNMq26cyzODOC4lOYjpY6kas61atigq
tachment%3B%20filename%3Dalertmanager-0.30.1.linux-amd64.tar.gz&response-content-type=application%2Foctet
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133,
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133
HTTP request sent, awaiting response... 200 OK
Length: 37196393 (35M) [application/octet-stream]
Saving to: 'alertmanager-0.30.1.linux-amd64.tar.gz'

alertmanager-0.30.1.linux-amd64.tar. 100%[=========================================================

2026-01-21 12:07:03 (273 MB/s) - 'alertmanager-0.30.1.linux-amd64.tar.gz' saved [37196393/37196393]

[root@ip-172-31-88-247 opt]#
```

- **tar xvf**

```
[root@ip-172-31-88-247 opt]# ls
alertmanager-0.30.1.linux-amd64.tar.gz  node_exporter-1.10.2.linux-amd64.tar
aws                                     prometheus-3.9.1.linux-amd64
node_exporter-1.10.2.linux-amd64        prometheus-3.9.1.linux-amd64.tar.gz
[root@ip-172-31-88-247 opt]# tar xvf alertmanager-0.30.1.linux-amd64.tar.gz
alertmanager-0.30.1.linux-amd64/
alertmanager-0.30.1.linux-amd64/alertmanager.yml
alertmanager-0.30.1.linux-amd64/NOTICE
alertmanager-0.30.1.linux-amd64/alertmanager
alertmanager-0.30.1.linux-amd64/LICENSE
alertmanager-0.30.1.linux-amd64/amtool
[root@ip-172-31-88-247 opt]#
```

Go to that alert manager by cd

```
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# ls
LICENSE  NOTICE  alertmanager  alertmanager.yml  amtool
```

- **vi alertmanager.yml**

```
global:
  smtp_smarthost: 'smtp.gmail.com:587'
  smtp_from: 'mujjus131@gmail.com'
  smtp_auth_username: 'mujjus131.com'
  smtp_auth_password: '987654321'
  smtp_require_tls: true

route:
  receiver: 'email-alerts'

receivers:
- name: 'email-alerts'
  email_configs:
  - to: 'mujjus896@gmail.com'
    send_resolved: true
```

Create a service file for alertmanager

- **vi /etc/systemd/system/alertmanager.service**

```
[Unit]
Description=Alertmanager Service
After=network.target

[Service]
Type=simple
ExecStart=/opt/alertmanager-0.30.1.linux-amd64/alertmanager \
  --config.file=/opt/alertmanager-0.30.1.linux-amd64/alertmanager.yml \
  --storage.path=/opt/alertmanager-0.30.1.linux-amd64/data
Restart=always

[Install]
WantedBy=multi-user.target
```

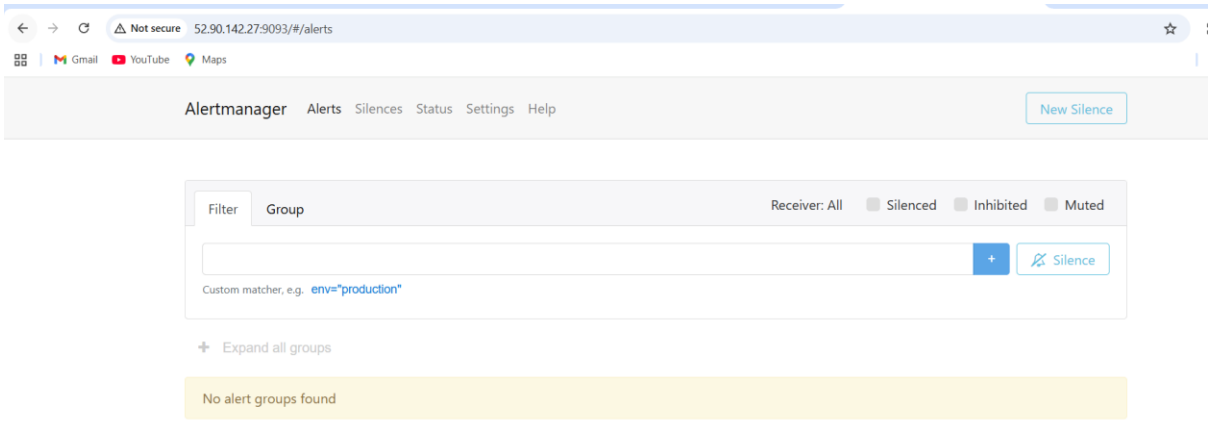
- **systemctl daemon-reload**
- **systemctl restart alertmanager**
- **systemctl status alertmanager**

```
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# vi /etc/systemd/system/alertmanager.service
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# vi /etc/systemd/system/alertmanager.service
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# systemctl daemon-reload
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# systemctl start alertmanager
[root@ip-172-31-88-247 alertmanager-0.30.1.linux-amd64]# systemctl status alertmanager
● alertmanager.service - Alertmanager Service
     Loaded: loaded (/etc/systemd/system/alertmanager.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 12:28:36 UTC; 11s ago
   Main PID: 7394 (alertmanager)
      Tasks: 8 (limit: 1067)
     Memory: 11.6M
        CPU: 63ms
     CGroup: /system.slice/alertmanager.service
             └─7394 /opt/alertmanager-0.30.1.linux-amd64/alertmanager --config.file=/opt/alertmanager-0.30.1.

Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.285Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.285Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.286Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.288Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.314Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.315Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.319Z level=INFO sou
Jan 21 12:28:36 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:36.319Z level=INFO sou
Jan 21 12:28:38 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:38.288Z level=INFO sou
Jan 21 12:28:46 ip-172-31-88-247.ec2.internal alertmanager[7394]: time=2026-01-21T12:28:46.291Z level=INFO sou
lines 1-20/20 (END)
```

Search with the ip address and port number 9093

You will see a web page of alerts manager



Connect alertmanager to prometheus

Go to Prometheus directory

- **vi Prometheus.yml**

add the port of alertmanager

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this
        labels:
          app: "prometheus"
```

Add this

- **- “localhost:9093”**

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
           - "localhost:9093"

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this
        labels:
          app: "prometheus"
~
```

- **systemctl restart prometheus**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# vi prometheus.yml
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl restart prometheus
```

Create a directory in Prometheus directory named as rules

- **mkdir rules**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ls
LICENSE  NOTICE  data  prometheus  prometheus.yml  promtool  rules
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# cd rules/
[root@ip-172-31-88-247 rules]# ls
[root@ip-172-31-88-247 rules]# vi alerts.yml
[root@ip-172-31-88-247 rules]#
```

go to rules directory and create a file

- **vi alerts.yml**

```yaml
groups:
- name: system-alerts
  rules:
  - alert: HighCPUUsage
    expr: 100 - (avg by(instance)(irate(node_cpu_seconds_total{mode="idle"}[2m])) * 100) > 80
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "High CPU Usage"
      description: "CPU usage is above 80% for more than 1 minute"

  - alert: HighMemoryUsage
    expr: (1 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes)) * 100 > 85
    for: 1m
    labels:
      severity: warning
    annotations:
      summary: "High Memory Usage"
      description: "Memory usage is above 85%"
```

Go to Prometheus.yml file add your alert file in the Prometheus file.

```yaml
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Defau
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          - "localhost:9093"

# Load rules once and periodically evaluate them according to the global 'e
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scr
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
       # The label name is added as a label `label_name=<label_value>` to a
        labels:
          app: "prometheus"
~
```

Add in rule_files

- **“- rules/ *.yml”**

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            - "localhost:9093"

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  - "rules/*.yml"
  # - "first_rules.yml"
  # - "second_rules.yml"

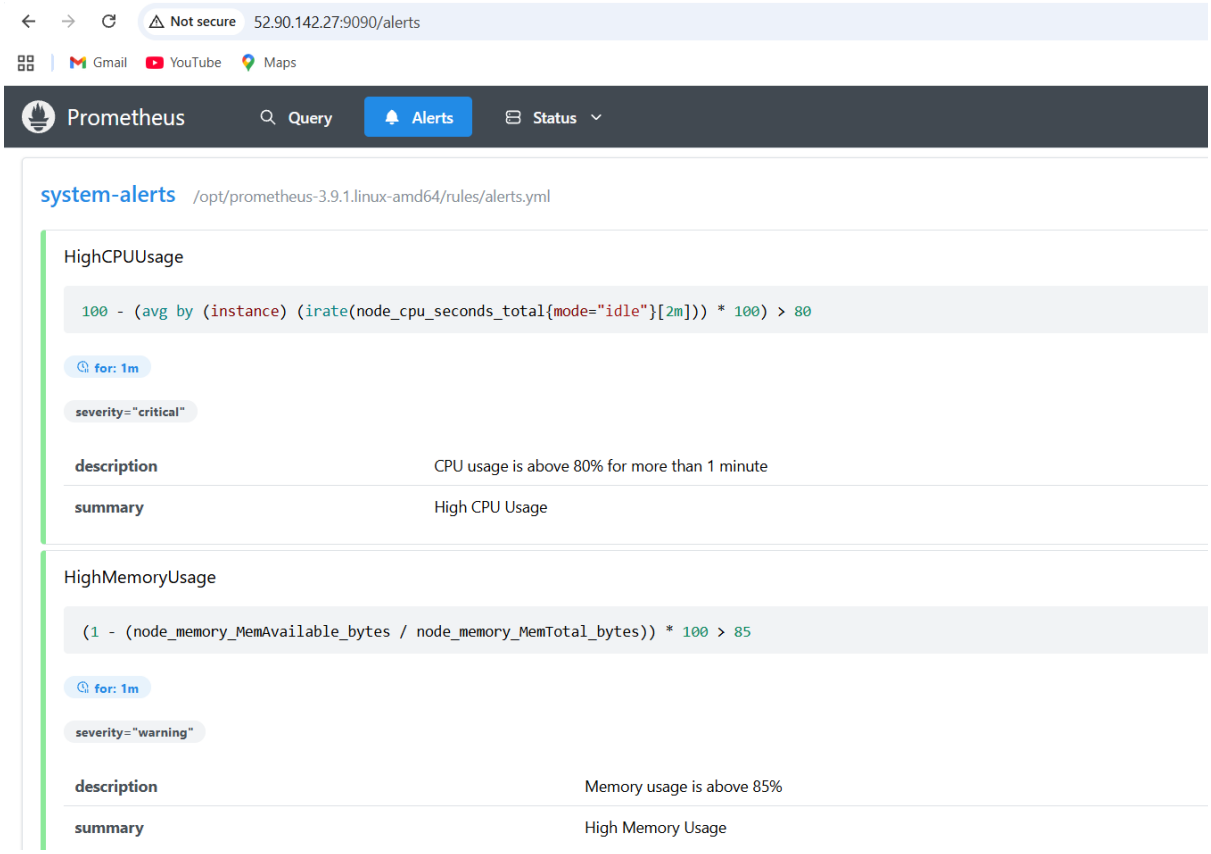
# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
        # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from th
```

You can test that alert by sending fake load to that server.

Search in browser with ip and port 9000:alerts



6. Install Grafana and add Prometheus as a data source.

Download Grafana by this commands in Prometheus server

- **wget https://dl.grafana.com/grafana-enterprise/release/12.3.1/grafana-enterprise_12.3.1_20271043721_linux_amd64.tar.gz**
- **tar -zxvf grafana-enterprise_12.3.1_20271043721_linux_amd64.tar.gz**

```
grafana-12.3.1/public/build/img/rendering_limit_light.png
grafana-12.3.1/public/build/img/grafana_icon.svg
grafana-12.3.1/public/build/img/grafanaconline.png
grafana-12.3.1/public/build/img/background_tease.jpg
grafana-12.3.1/public/build/img/grab_dark.svg
grafana-12.3.1/public/build/img/browserconfig.xml
grafana-12.3.1/public/build/img/icn-renderer.svg
grafana-12.3.1/public/build/img/grafana_mask_icon.svg
grafana-12.3.1/public/build/img/icn-dashboard-tiny.svg
grafana-12.3.1/public/build/img/grafana_net_logo.svg
grafana-12.3.1/public/build/img/checkbox.png
grafana-12.3.1/public/build/TeamPages.7ef24e1eb009e3f781a7.js
grafana-12.3.1/public/build/3466.fda7da0cb4150e59da98.js.map
grafana-12.3.1/public/build/2150.7576285186c713978b39.js
grafana-12.3.1/public/build/6990.d663924714fc213b17c6.js.map
grafana-12.3.1/public/build/AlertGroups.81264e6e5009c36eed6e.js
grafana-12.3.1/public/build/9355.95d7e1782f93f339549e.js.map
grafana-12.3.1/public/build/ConnectionsDataSourcePermissionsUpgradePage.9831a94a4822f990e8b5.js.map
grafana-12.3.1/public/build/ContactPoints.9ae2d1927d60c6169b87.js.map
grafana-12.3.1/public/build/8018.c31846109f2dff21d8c4.js.map
grafana-12.3.1/public/build/loki-query-field.f9a0b82b636d4821760c.js.map
grafana-12.3.1/public/build/9150.38487365ffd149b730af.js
grafana-12.3.1/public/build/3630.1ffe9229d5c27fd0823a.js.map
grafana-12.3.1/public/build/radialBarPanel.294c52d497d8f5c2bebb.js.map
grafana-12.3.1/public/build/2710.e1ac8e57578a838f5bd8.js.map
grafana-12.3.1/public/build/7806.dff3509790e494ca5a45.js
grafana-12.3.1/plugins-bundled/
[root@ip-172-31-88-247 opt]# ls
alertmanager-0.30.1.linux-amd64                      node_exporter-1.10.2.linux-amd64.tar.gz
alertmanager-0.30.1.linux-amd64.tar.gz               prometheus-3.9.1.linux-amd64
aws                                                  prometheus-3.9.1.linux-amd64.tar.gz
grafana-12.3.1                                       snmp_exporter-0.25.0.linux-amd64
grafana-enterprise_12.3.1_20271043721_linux_amd64.tar.gz  snmp_exporter-0.25.0.linux-amd64.tar.gz
node_exporter-1.10.2.linux-amd64
[root@ip-172-31-88-247 opt]#
```

- **cd grafana**
- **cd bin**

```
[root@ip-172-31-88-247 opt]# cd grafana-12.3.1/
[root@ip-172-31-88-247 grafana-12.3.1]# ls
Dockerfile  LICENSE  NOTICE.md  README.md  VERSION  bin  conf
[root@ip-172-31-88-247 grafana-12.3.1]# cd bin
[root@ip-172-31-88-247 bin]# ls
grafana  grafana-cli  grafana-server
[root@ip-172-31-88-247 bin]#
```

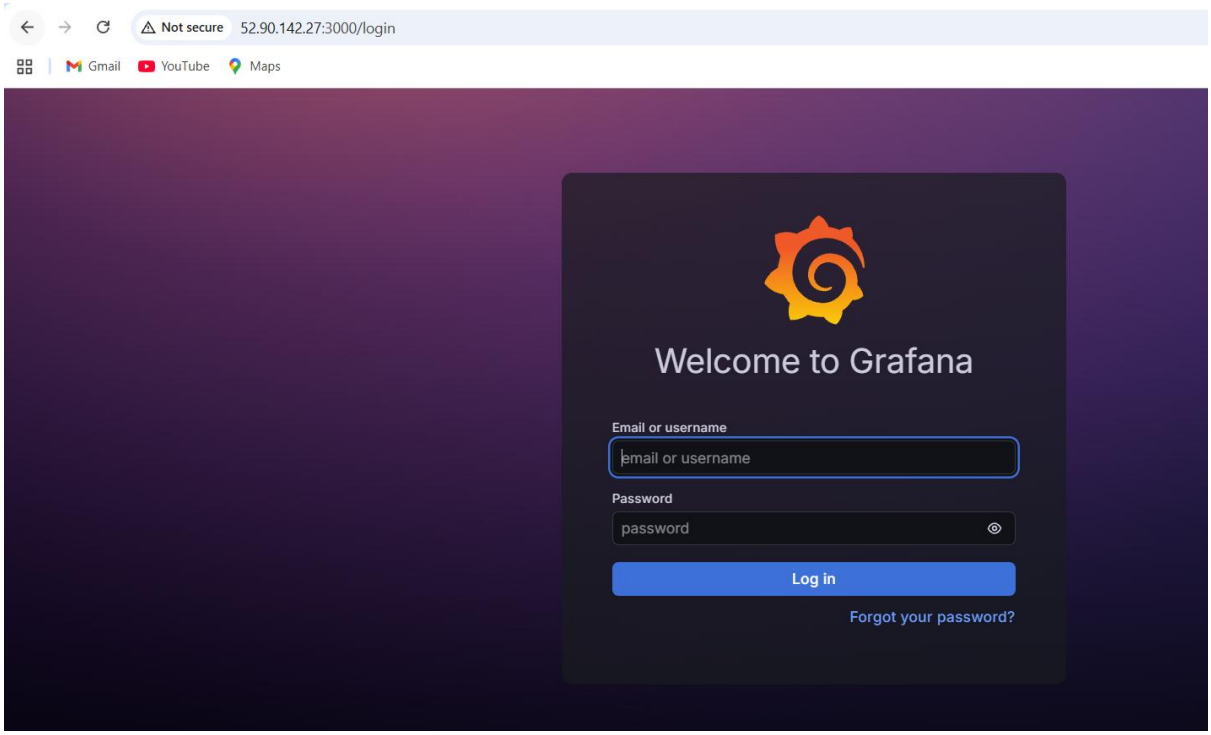
- **./grafana-server**

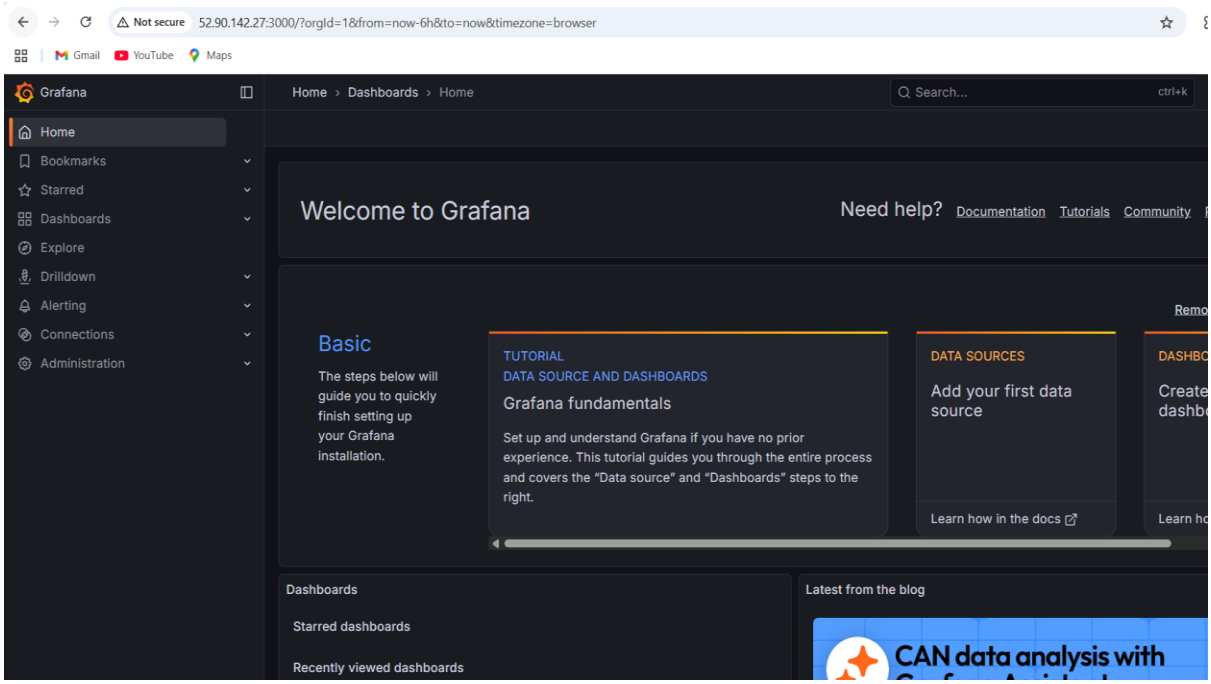
```
[root@ip-172-31-88-247 bin]# ls
grafana  grafana-cli  grafana-server
[root@ip-172-31-88-247 bin]# ./grafana-server
Deprecation warning: The standalone 'grafana-server' program is deprecated and will be removed in th
 'grafana-server' to 'grafana server'
Grafana server is running with elevated privileges. This is not recommended
INFO [01-21|13:49:01] Starting Grafana                         logger=settings version=12.3.1 commit
247fba branch=HEAD compiled=2026-01-21T13:49:01Z
INFO [01-21|13:49:01] Config loaded from                       logger=settings file=/opt/grafana-12.
INFO [01-21|13:49:01] Target                                   logger=settings target=[all]
INFO [01-21|13:49:01] Path Home                                logger=settings path=/opt/grafana-12.
INFO [01-21|13:49:01] Path Data                                logger=settings path=/opt/grafana-12.
INFO [01-21|13:49:01] Path Logs                                logger=settings path=/opt/grafana-12.
INFO [01-21|13:49:01] Path Plugins                             logger=settings path=/opt/grafana-12.
INFO [01-21|13:49:01] Path Provisioning                        logger=settings path=/opt/grafana-12.
INFO [01-21|13:49:01] App mode production                      logger=settings
INFO [01-21|13:49:01] FeatureToggles                           logger=featuremgmt awsAsyncQueryCachi
lingSAML=true logRowsPopoverMenu=true alertRuleRestore=true alertingSaveStateCompressed=true lokiQue
rrideAudience=true grafanaAssistantInProfilesDrilldown=true dashboardScene=true alertingUIOptimizeRe
ring=true alertingMigrationUI=true kubernetesDashboards=true cloudWatchRoundUpEndTime=true dataplane
nsRedesign=true alertingUseNewSimplifiedRoutingHashAlgorithm=true preinstallAutoUpdate=true cloudWat
ossAccountQuerying=true dashboardDsAdHocFiltering=true onlyStoreActionSets=true formatString=true ne
=true newLogsPanel=true alertingRulePermanentlyDelete=true useSessionStorageForRedirection=true aler
ersUI=true alertingNotificationsStepMode=true azureMonitorPrometheusExemplars=true addFieldFromCalcu
sLDAP=true recordedQueriesMulti=true azureMonitorEnableUserAuth=true alertingQueryAndExpressionsStep
true preventPanelChromeOverflow=true sharingDashboardImage=true dashboardSceneForViewers=true unifie
ourcesTempCredentials=true logsContextDatasourceUi=true onPremToCloudMigrations=true dashboardScenes
```

Search in browser with the ip and port number 3000

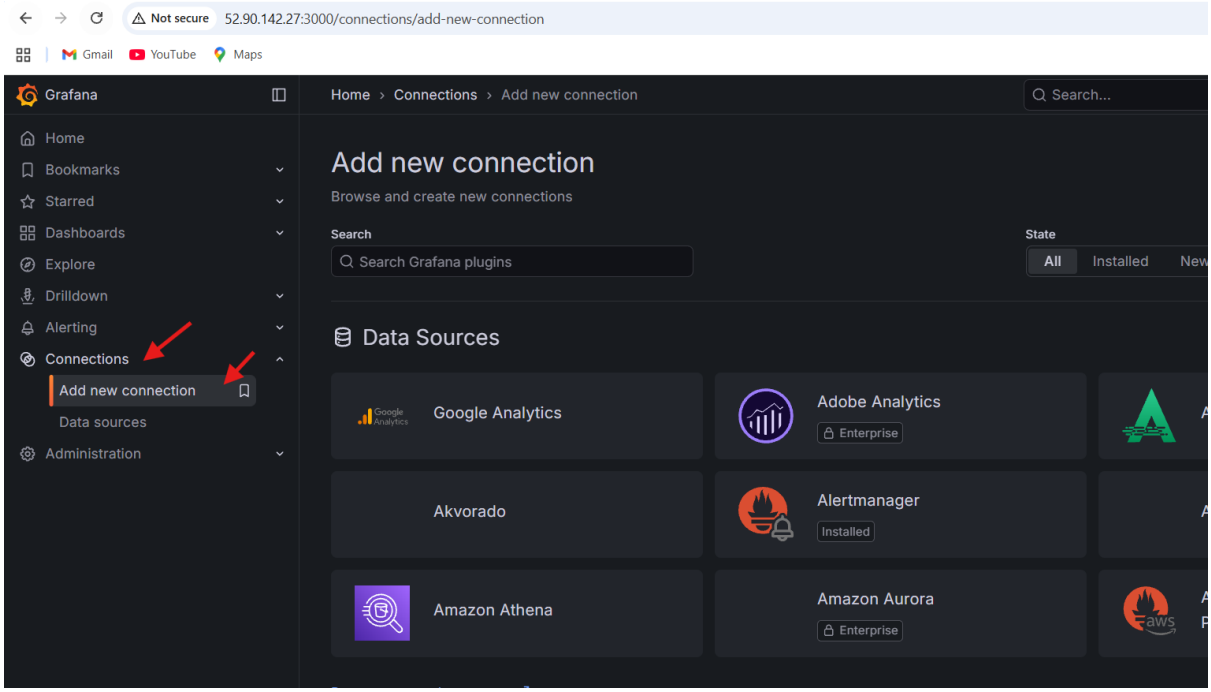
Username:admin

Password:admin

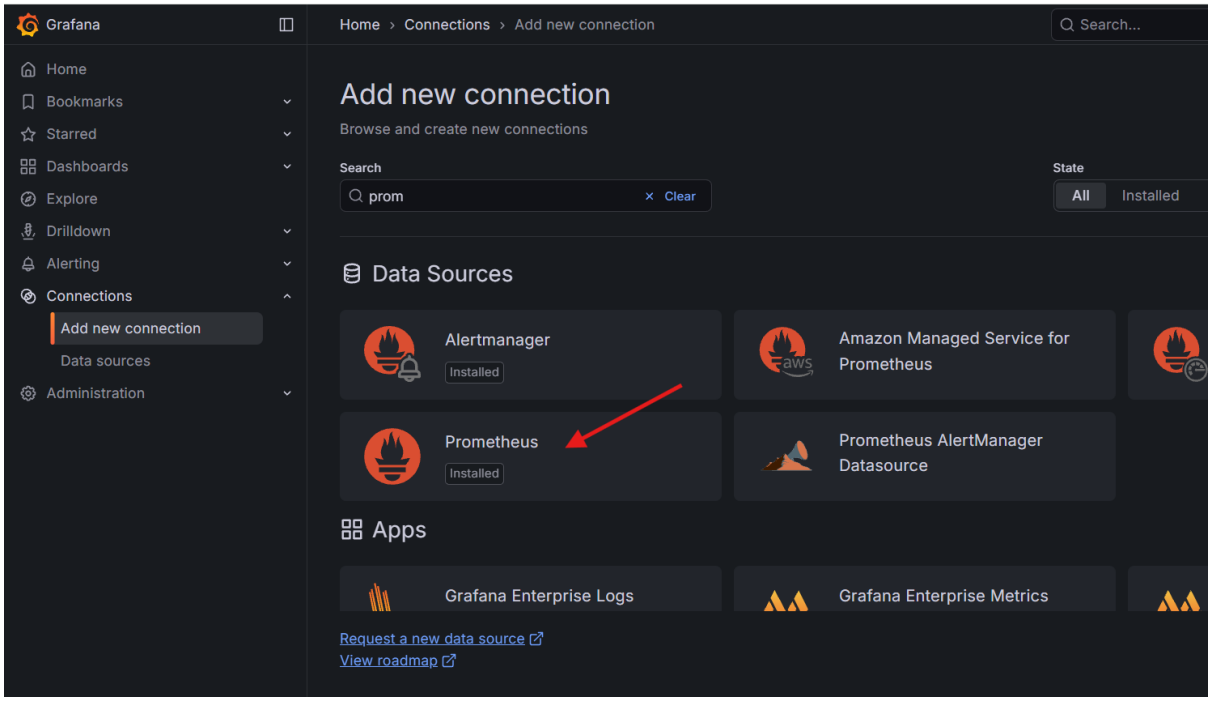




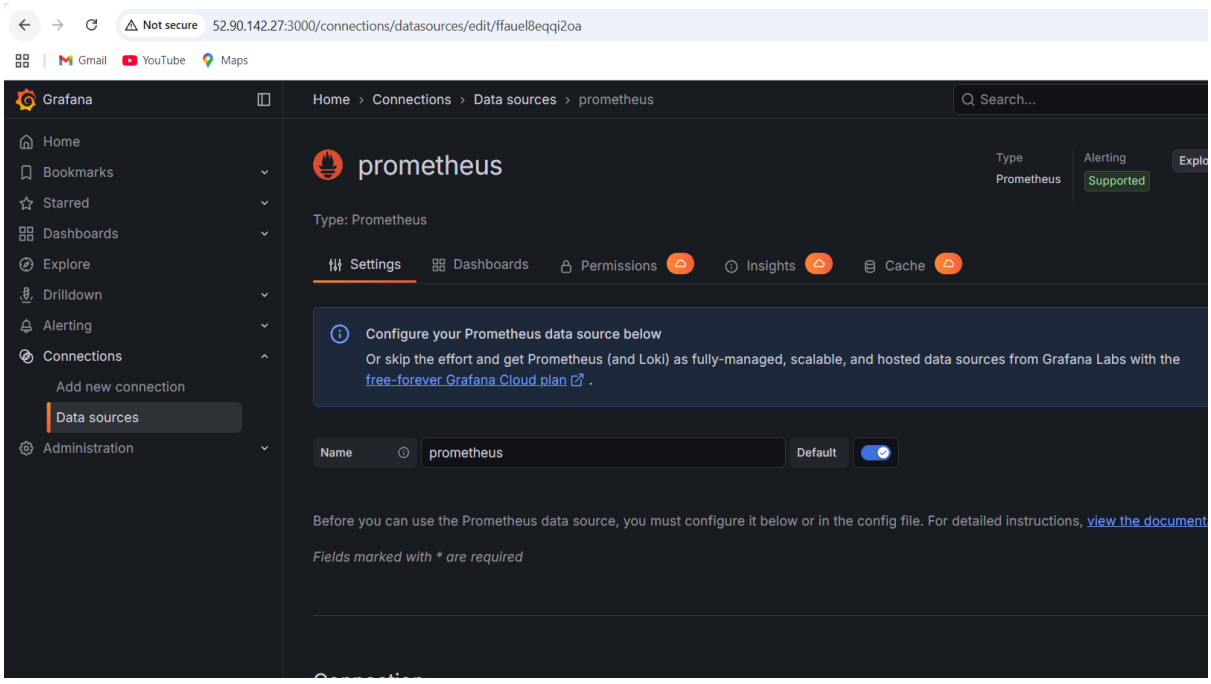
Click on connection, add new connection



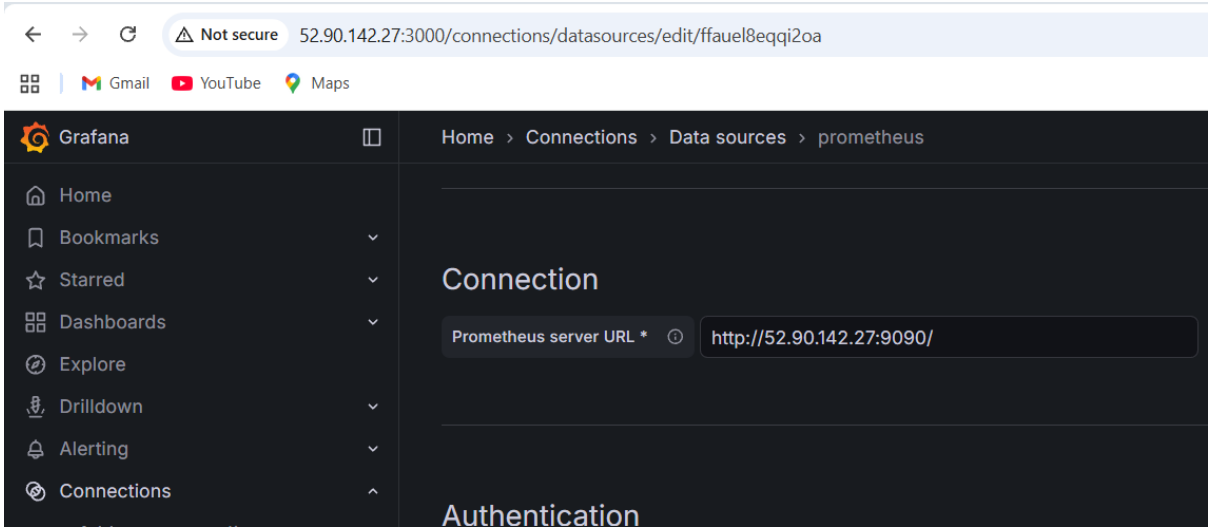
Search for Prometheus



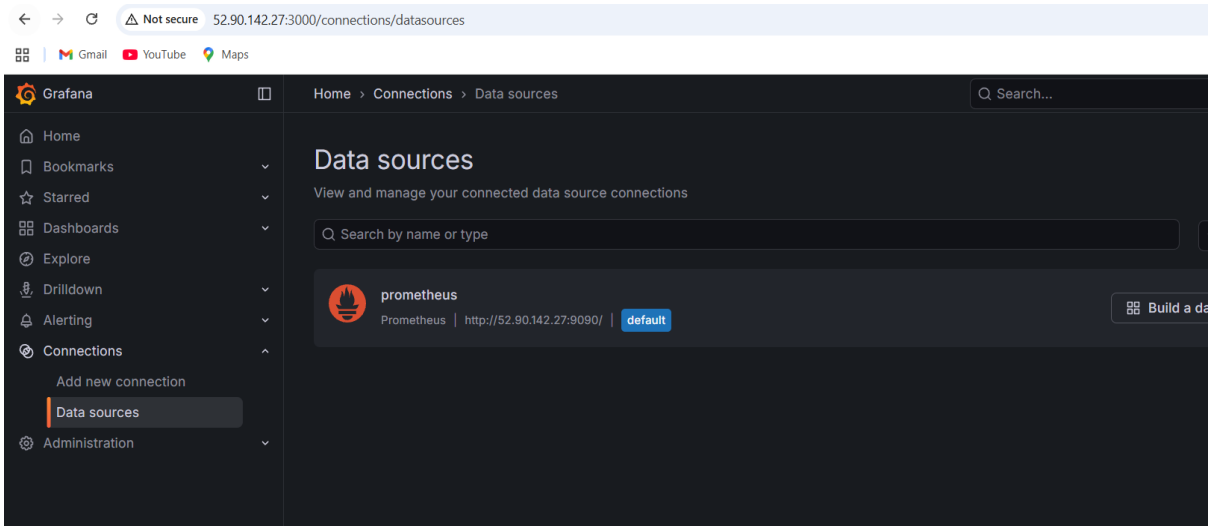
Add new data source



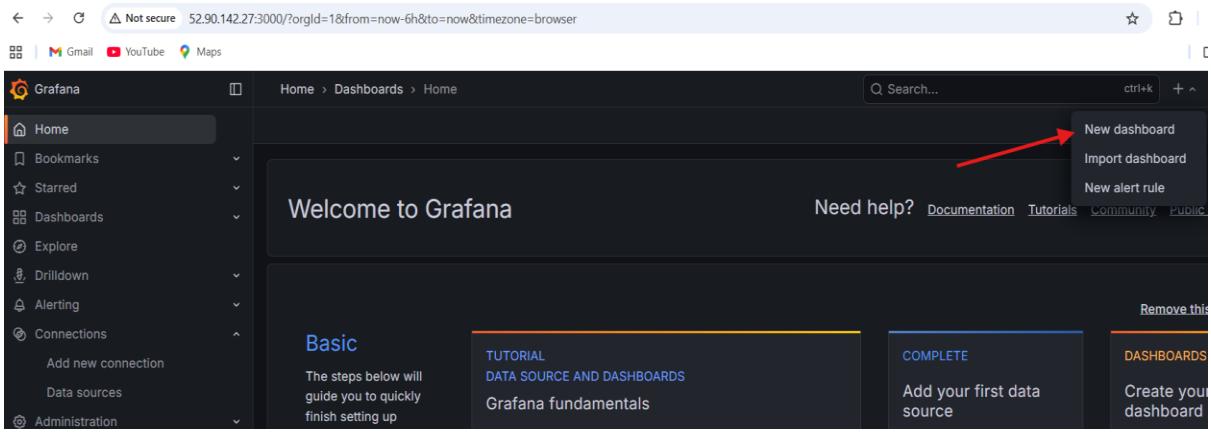
Pass your ip address and give there

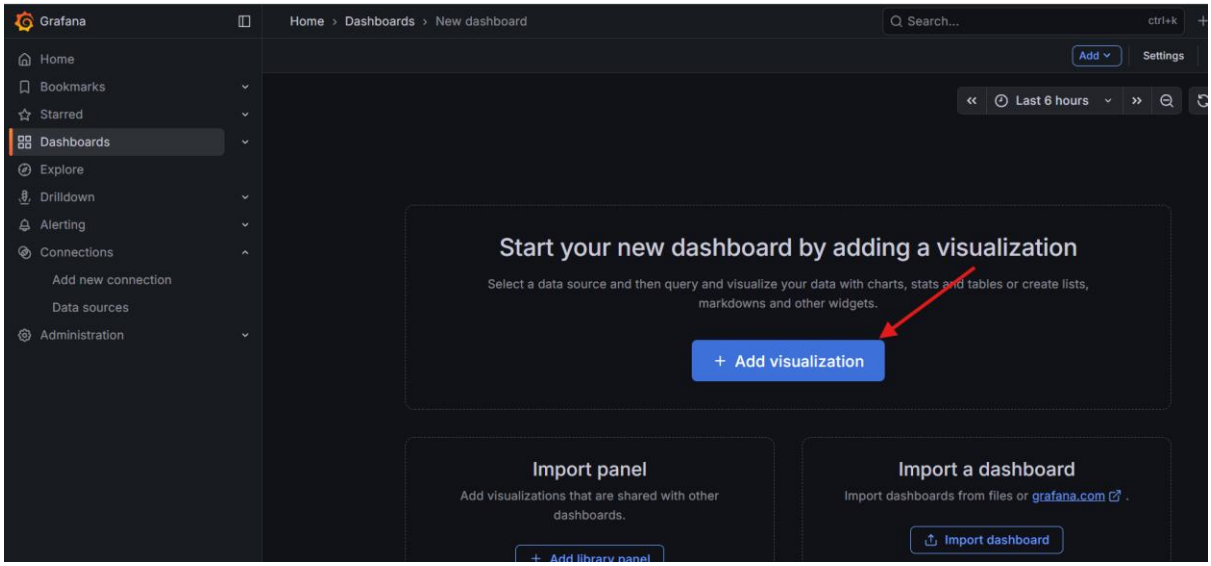


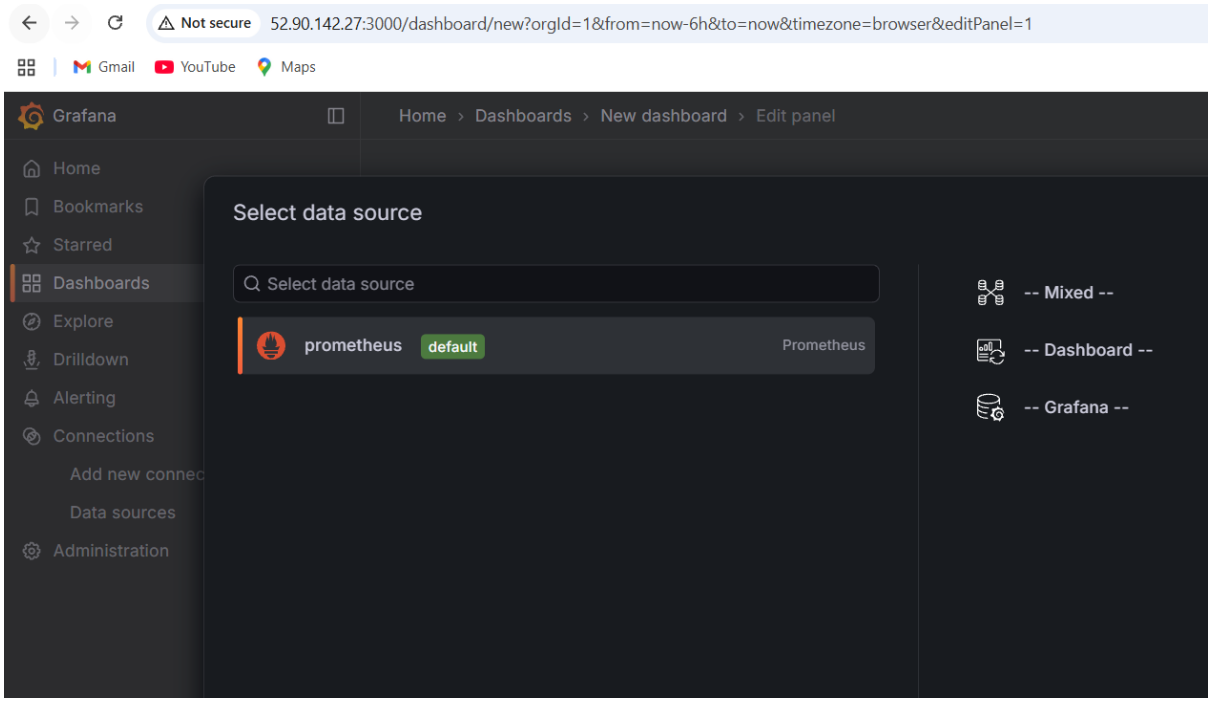
Click on test and save then you will have data source of Prometheus.



Create new dashboard

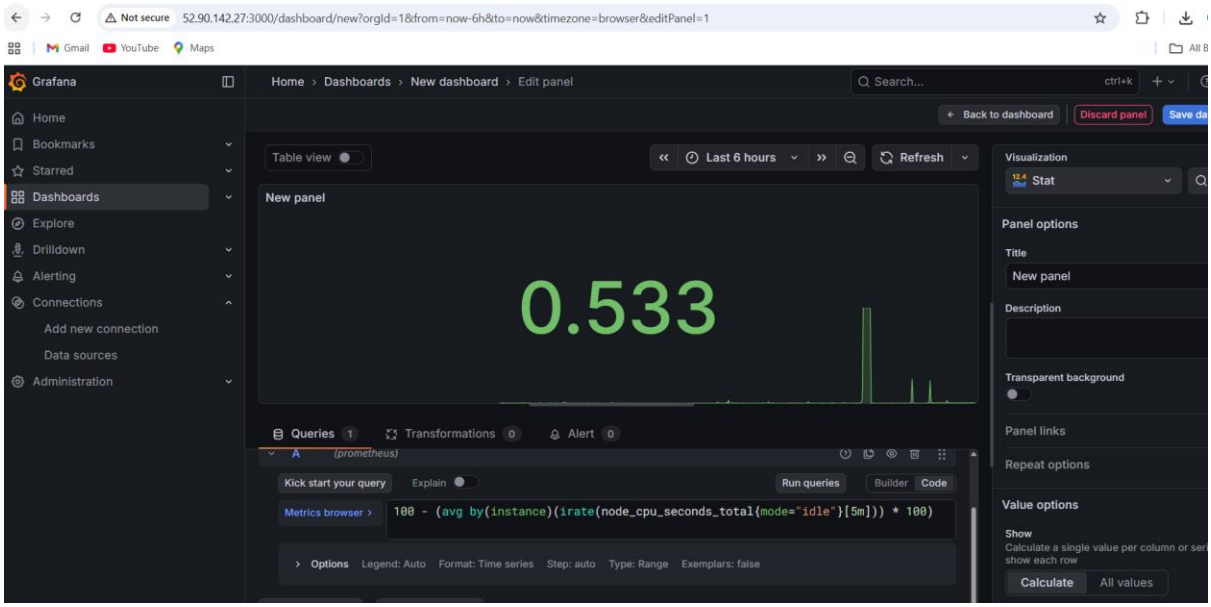






Add the query which you want like cpu, uptime…

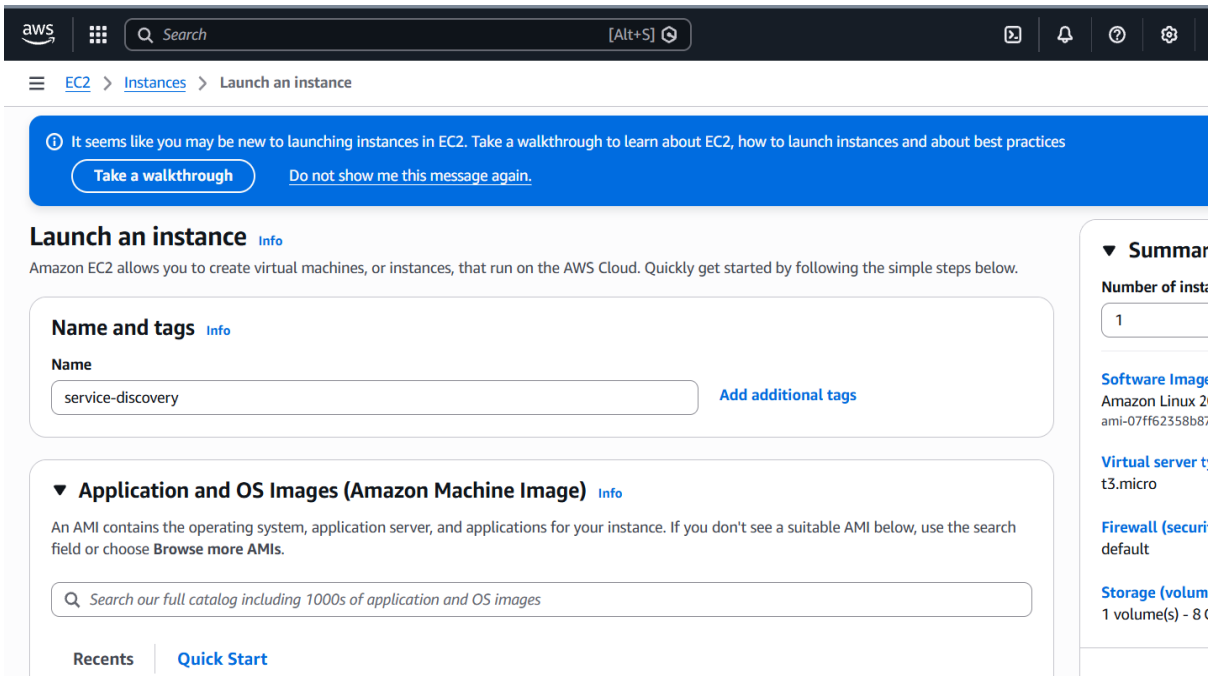
Click on run queries



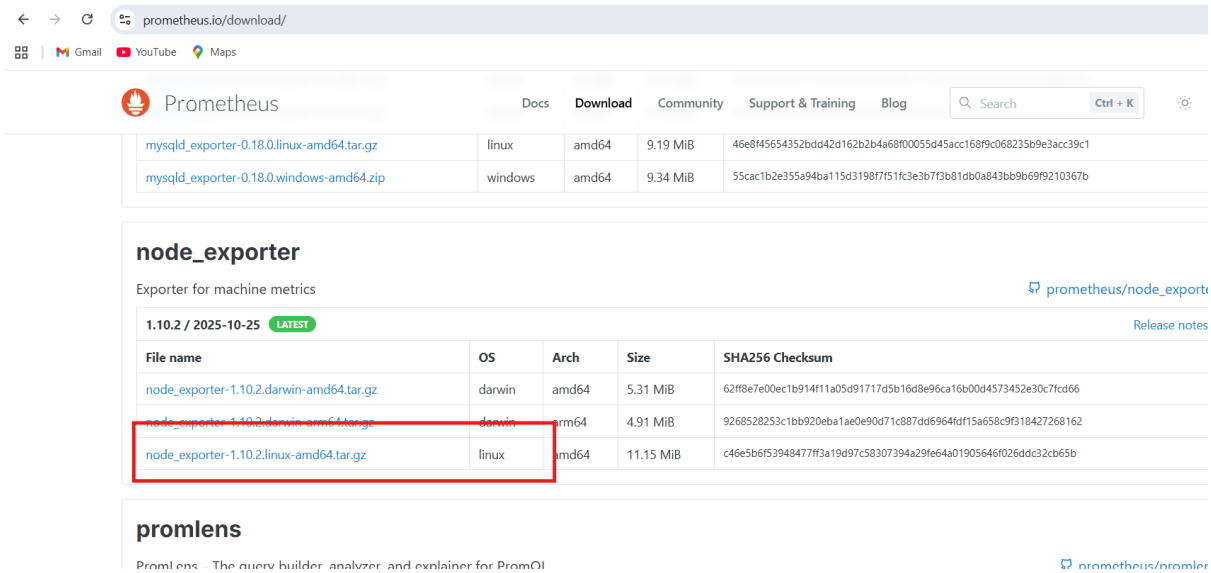
7. Configure Prometheus for service discovery of dynamic targets.

Manual:

Create ec2-instance with the name service discovery



Copy the link of node exporter and wget in the target machine




```
[root@ip-172-31-107-144 ~]# wget https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node
--2026-01-21 16:17:46--  https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node_exporte
Resolving github.com (github.com)... 140.82.114.3
Connecting to github.com (github.com)|140.82.114.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/23923325-96f
2026-01-21T16%3A59%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&rsct=applica
&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-21T15%3A58%3A47Z&ske=2026-01-21T16%3A59%3A15Z&sks=b
nt0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ21
DA2NiwibmJmIjoxNzY5MDEyMjY2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.rC6iS
ition=attachment%3B%20filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&response-content-type=application%
--2026-01-21 16:17:46--  https://release-assets.githubusercontent.com/github-production-release-asset/95240
b&spr=https&se=2026-01-21T16%3A59%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.10.2.linux-amd64.tar.
dd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-21T15%3A58%3A47Z&ske=2026-01-21T16%3
xrp6IUztwrMwNn0nt0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFz
V4cCI6MTc2OTAxNDA2NiwibmJmIjoxNzY5MDEyMjY2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy
-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&response-content-ty
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 1
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:
HTTP request sent, awaiting response... 200 OK
Length: 11686705 (11M) [application/octet-stream]
Saving to: 'node_exporter-1.10.2.linux-amd64.tar.gz'

node_exporter-1.10.2.linux-amd64.tar.gz    100%[=========================================================
```

- **tar xvf nodeport**


```
[root@ip-172-31-107-144 ~]# wget https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node
--2026-01-21 16:17:46--  https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node_exporte
Resolving github.com (github.com)... 140.82.114.3
Connecting to github.com (github.com)|140.82.114.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/23923325-96f
2026-01-21T16%3A59%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&rsct=applica
&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-21T15%3A58%3A47Z&ske=2026-01-21T16%3A59%3A15Z&sks=b
nt0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ21
DA2NiwibmJmIjoxNzY5MDEyMjY2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.rC6iS
ition=attachment%3B%20filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&response-content-type=application%
--2026-01-21 16:17:46--  https://release-assets.githubusercontent.com/github-production-release-asset/95240
b&spr=https&se=2026-01-21T16%3A59%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.10.2.linux-amd64.tar.
dd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-01-21T15%3A58%3A47Z&ske=2026-01-21T16%3
xrp6IUztwrMwNn0nt0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFz
V4cCI6MTc2OTAxNDA2NiwibmJmIjoxNzY5MDEyMjY2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy
-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.10.2.linux-amd64.tar.gz&response-content-ty
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 1
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:
HTTP request sent, awaiting response... 200 OK
Length: 11686705 (11M) [application/octet-stream]
Saving to: 'node_exporter-1.10.2.linux-amd64.tar.gz'

node_exporter-1.10.2.linux-amd64.tar.gz    100%[=========================================================
```

```
[root@ip-172-31-107-144 ~]# tar xvf node_exporter-1.10.2.linux-amd
node_exporter-1.10.2.linux-amd64/
node_exporter-1.10.2.linux-amd64/node_exporter
node_exporter-1.10.2.linux-amd64/LICENSE
node_exporter-1.10.2.linux-amd64/NOTICE
[root@ip-172-31-107-144 ~]#
```

Create a service file

- **vi /etc/systemd/system/node_exporter.service**




```
[Unit]
Description=Node Exporter
After=network.target
[Service]
User=nobody
ExecStart=/opt/node_exporter-1.9.1.linux-amd64/node_exporter
Restart=on-failure
[Install]
WantedBy=multi-user. Target
~
```

- **systemctl daemon-reload**
- **systemctl restart node_exporter**
- **systemctl status node_exporter**

```
[root@ip-172-31-107-144 opt]# vi /etc/systemd/system/node_exporter.se
[root@ip-172-31-107-144 opt]# vi /etc/systemd/system/node_exporter.se
[root@ip-172-31-107-144 opt]# systemctl daemon-reload
[root@ip-172-31-107-144 opt]# systemctl restart node_exporter
[root@ip-172-31-107-144 opt]# systemctl status node_exporter
● node_exporter.service - Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; disab
     Active: active (running) since Wed 2026-01-21 16:34:00 UTC; 11s
   Main PID: 2658 (node_exporter)
      Tasks: 4 (limit: 1067)
     Memory: 2.6M
        CPU: 8ms
     CGroup: /system.slice/node_exporter.service
             └─2658 /opt/node_exporter-1.10.2.linux-amd64/node_export

Jan 21 16:34:00 ip-172-31-107-144.ec2.internal node_exporter[2658]: t
```

Connect discovery file to Prometheus server.

Go to Prometheus service and add the target ip address of your discovery service file.

```
[root@ip-172-31-88-247 ~]# cd /opt
[root@ip-172-31-88-247 opt]# ls
alertmanager-0.30.1.linux-amd64                          node_exporter-1.10.2.linux-amd64.tar.gz
alertmanager-0.30.1.linux-amd64.tar.gz                   prometheus-3.9.1.linux-amd64
aws                                                      prometheus-3.9.1.linux-amd64.tar.gz
grafana-12.3.1                                           snmp_exporter-0.25.0.linux-amd64
grafana-enterprise_12.3.1_20271043721_linux_amd64.tar.gz snmp_exporter-0.25.0.linux-amd64.tar.gz
node_exporter-1.10.2.linux-amd64
[root@ip-172-31-88-247 opt]# cd prometheus-3.9.1.linux-amd64/
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# ls
LICENSE  NOTICE  data  disk_alerts.yml  prometheus  prometheus.yml  promtool  rules
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# vi prometheus.yml
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]#
```

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 min
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            - "localhost:9093"

# Load rules once and periodically evaluate them according to the global 'evaluation_interva
rule_files:
  - "rules/*.yml"
  - "disk_alerts.yml"

  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this co
  - job_name: "prometheus"

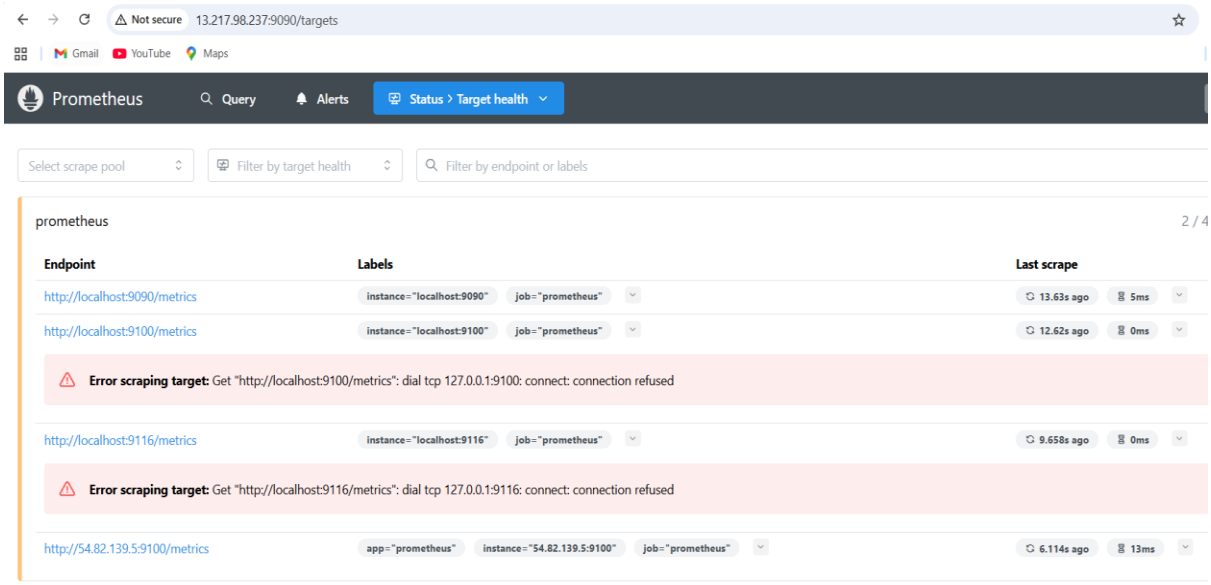
    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
      - targets: ["54.82.139.5:9100"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries scr
        labels:
```

- **systemctl restart Prometheus**

go to gui and see in targets your instance in the targets.

(Error scarping target will be there it is because of you didn’t installed node_exporter plugin or service file for node_exporter in that machine If you execute that plugin or service file then it will be up in state and error will gone.)



Service discovery:

Go to Prometheus server and add this

```
          - localhost:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_inter
rule_files:
  - "rules/*.yml"
  - "disk_alerts.yml"

  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
      - targets: ["54.82.139.5:9100"]
       # The label name is added as a label `label_name=<label_value>` to any timeseries s
        labels:
          app: "prometheus"

  - job_name: 'ec2-node-exporter'
    ec2_sd_configs:
      - region: us-east-1         # Change to your AWS region
        port: 9100                 # Node Exporter port
    relabel_configs:
      - source_labels: [__meta_ec2_tag_Name]
        regex: "prometheus-node.*"
        action: keep
```

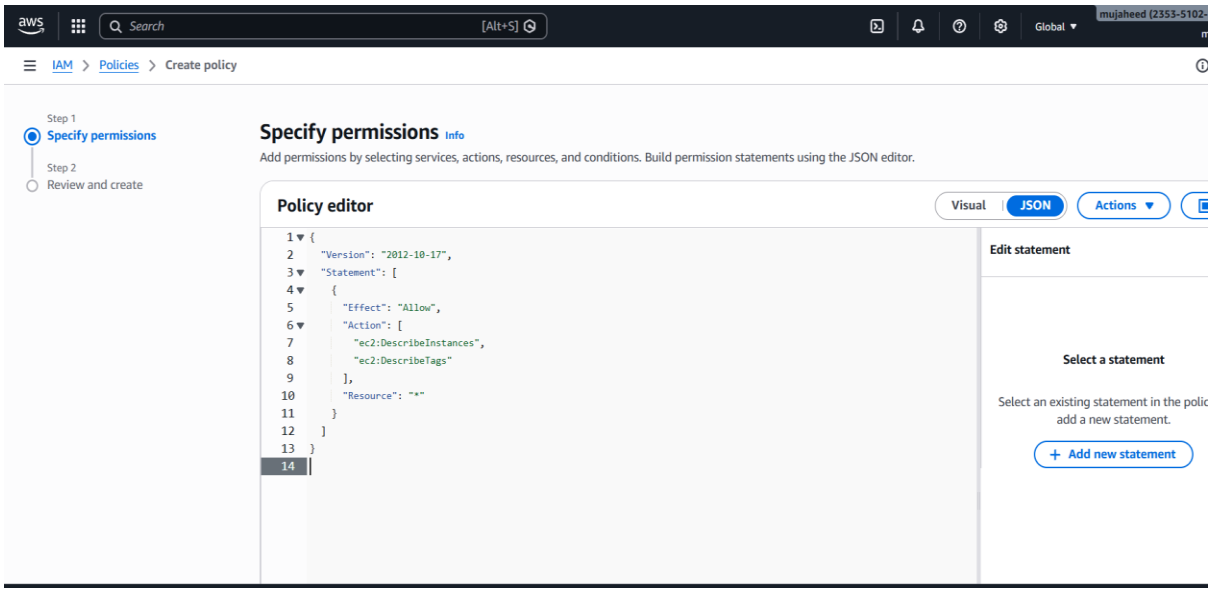
- **systemctl restart prometheus.service**
- **systemctl status prometheus.service**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl status prometheus.service
● prometheus.service - Prometheus Monitoring
     Loaded: loaded (/etc/systemd/system/prometheus.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 17:07:21 UTC; 2min 43s ago
   Main PID: 3896 (prometheus)
      Tasks: 8 (limit: 1067)
     Memory: 29.0M
        CPU: 269ms
     CGroup: /system.slice/prometheus.service
             └─3896 /opt/prometheus-3.9.1.linux-amd64/prometheus --config.file=/opt/prometheus-

Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.212Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.219Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.219Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.219Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.223Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.223Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.223Z
Jan 21 17:07:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:07:21.245Z
Jan 21 17:08:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:08:21.248Z
Jan 21 17:09:21 ip-172-31-88-247.ec2.internal prometheus[3896]: time=2026-01-21T17:09:21.247Z
```

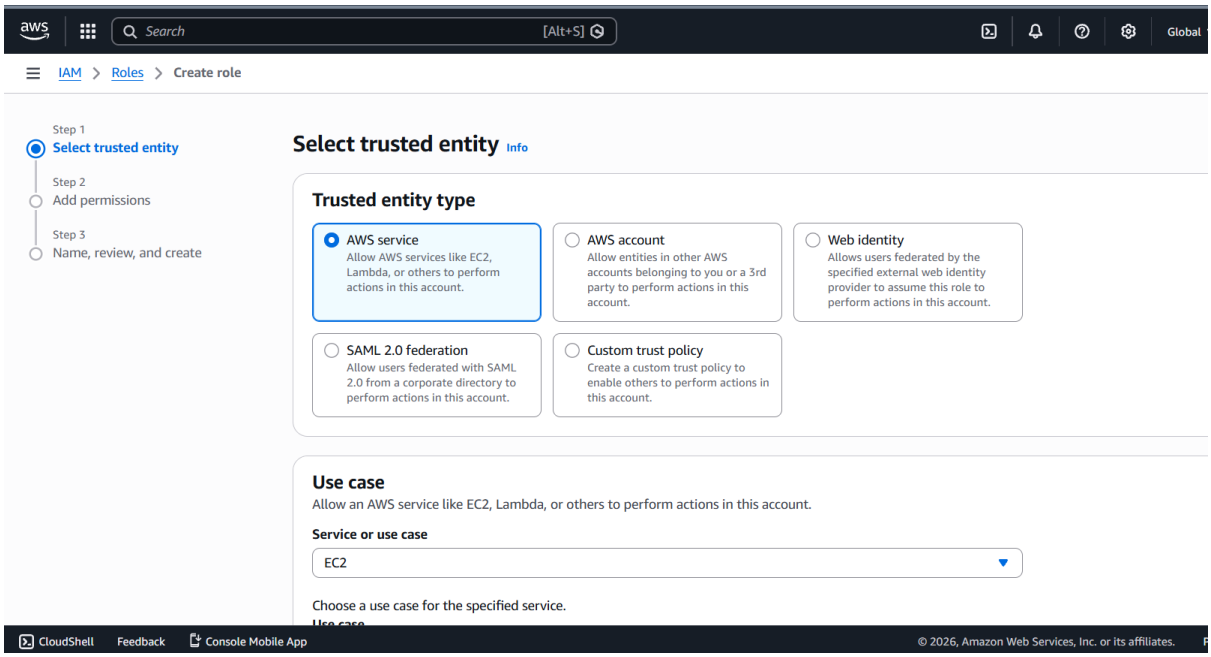
Go to IAM and create a policy with this code name as test-prometheus

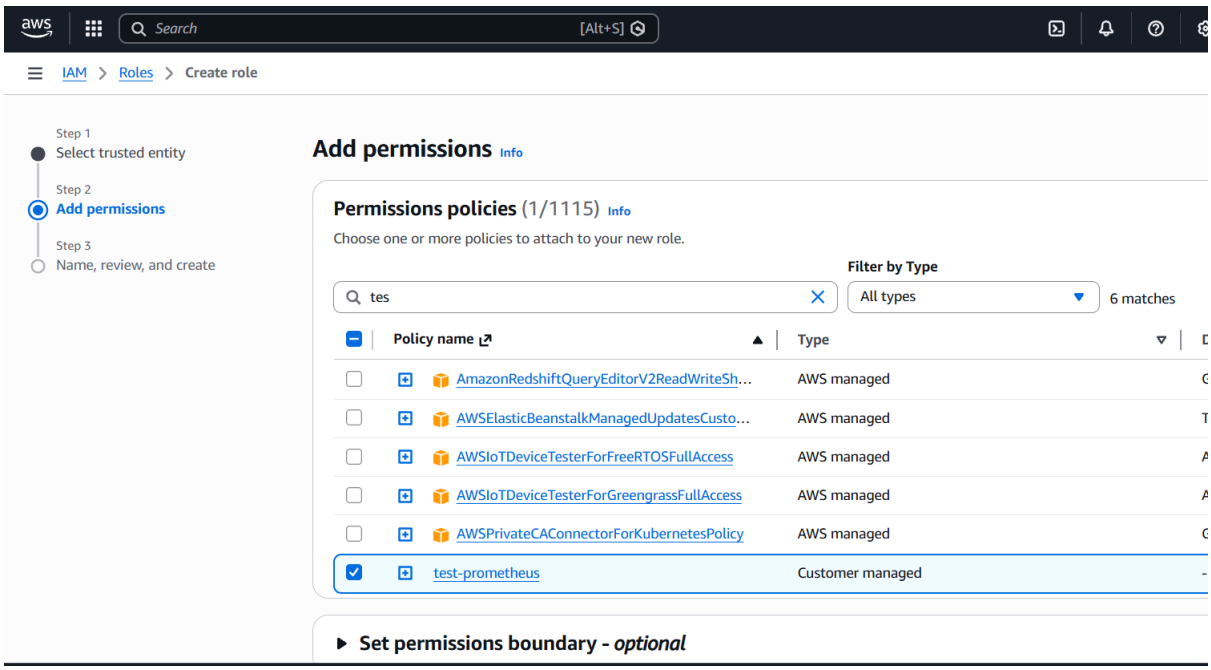
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "ec2:DescribeTags"
      ],
      "Resource": "*"
    }
  ]
}
```

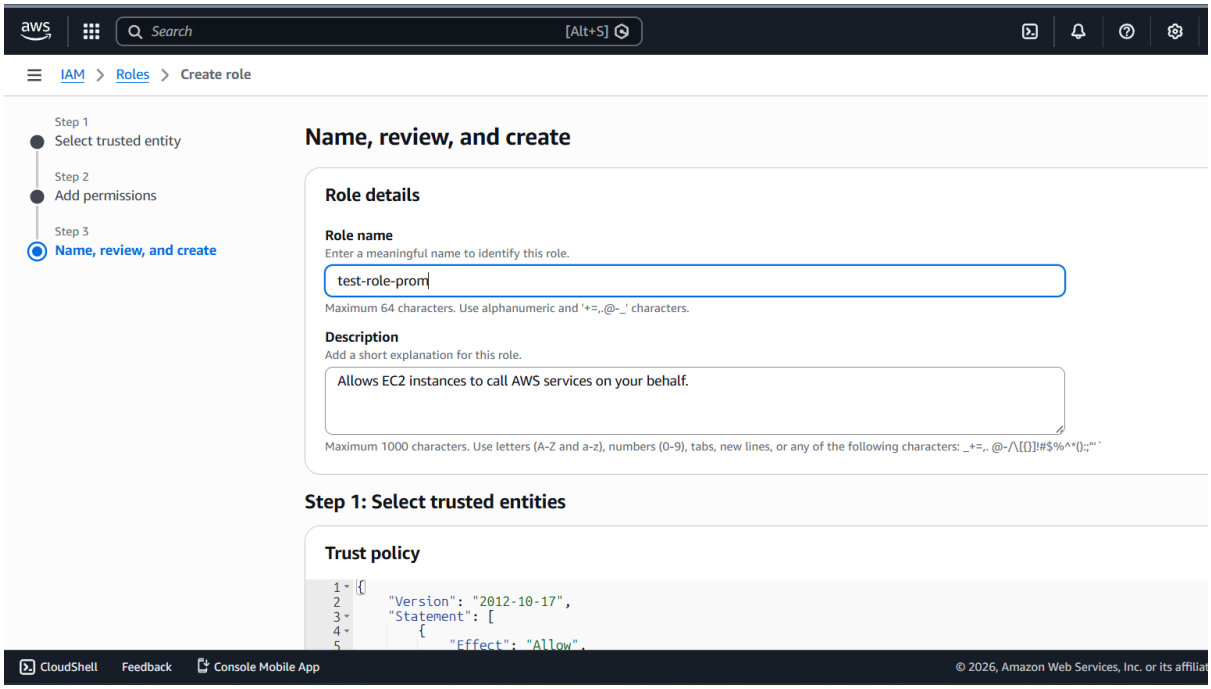


Attach the policy to the role

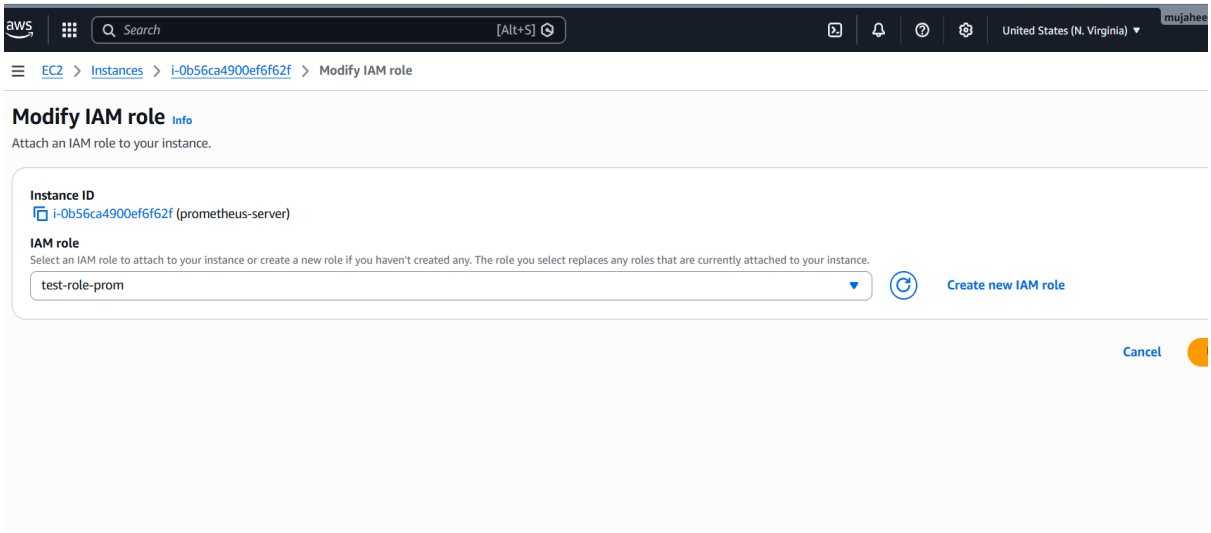
Create a role





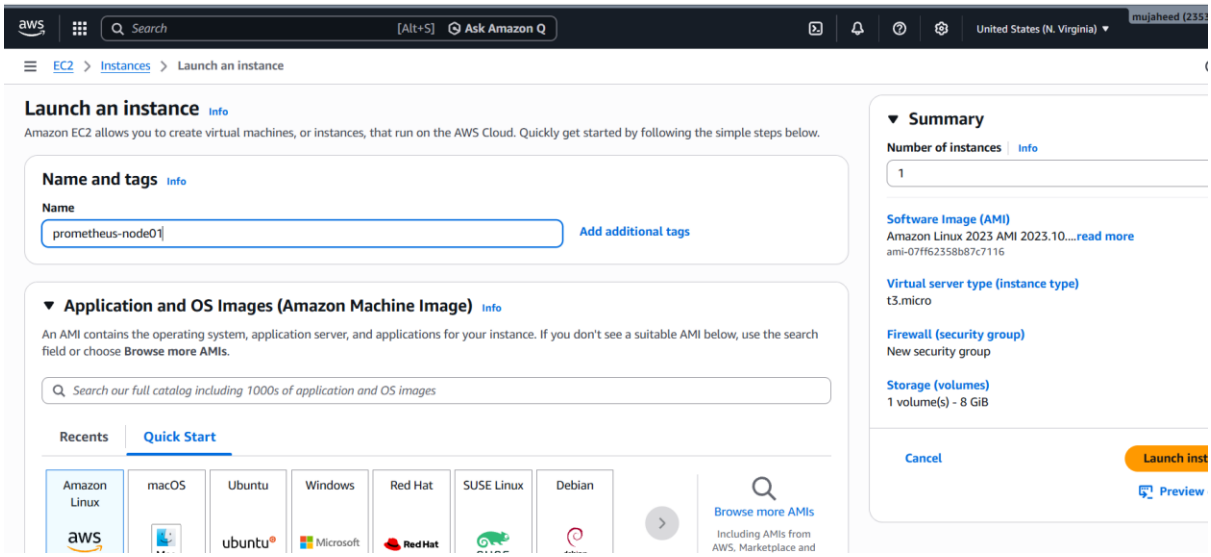


Attach the created role to Prometheus server

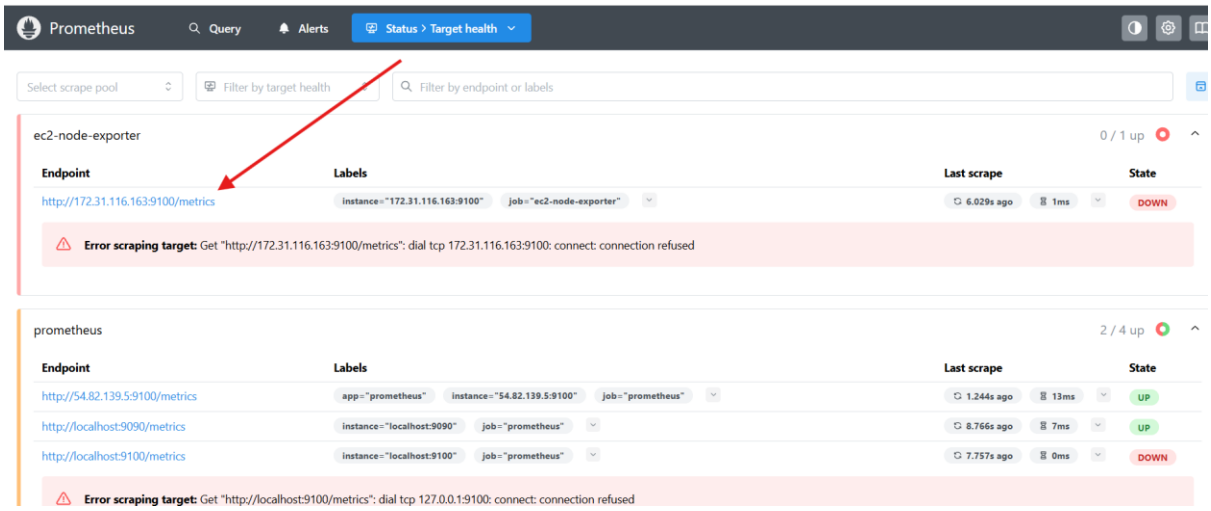


If you create a new instance with the name Prometheus-node

(It was a label that you provided in the script)then it will automatically monitor into your target health in Prometheus server.



Go to Prometheus gui and refresh you will see a end point has been created automatically with your ip address.



Error scarping target will be there it is because of you didn’t installed node_exporter plugin or service file for node_exporter in that machine If you execute that plugin or service file then it will be up in state and error will gone.

8. Create alerting rules for monitoring disk space.

Go to Prometheus server and create a file disk_alerts.yml

```
groups:
- name: disk_alerts
  rules:
  - alert: DiskSpaceLow
    expr:
(node_filesystem_avail_bytes{fstype!~"tmpfs|overlay"} /
node_filesystem_size_bytes{fstype!~"tmpfs|overlay"}) *
100 < 15
    for: 2m
    labels:
      severity: warning
    annotations:
      summary: "Low disk space on {{ $labels.instance }}"
      description: "Disk available is below 15% on {{
$labels.mountpoint }} (instance {{ $labels.instance }})"

  - alert: DiskSpaceCritical
    expr:
(node_filesystem_avail_bytes{fstype!~"tmpfs|overlay"} /
node_filesystem_size_bytes{fstype!~"tmpfs|overlay"}) *
100 < 5
    for: 1m
```

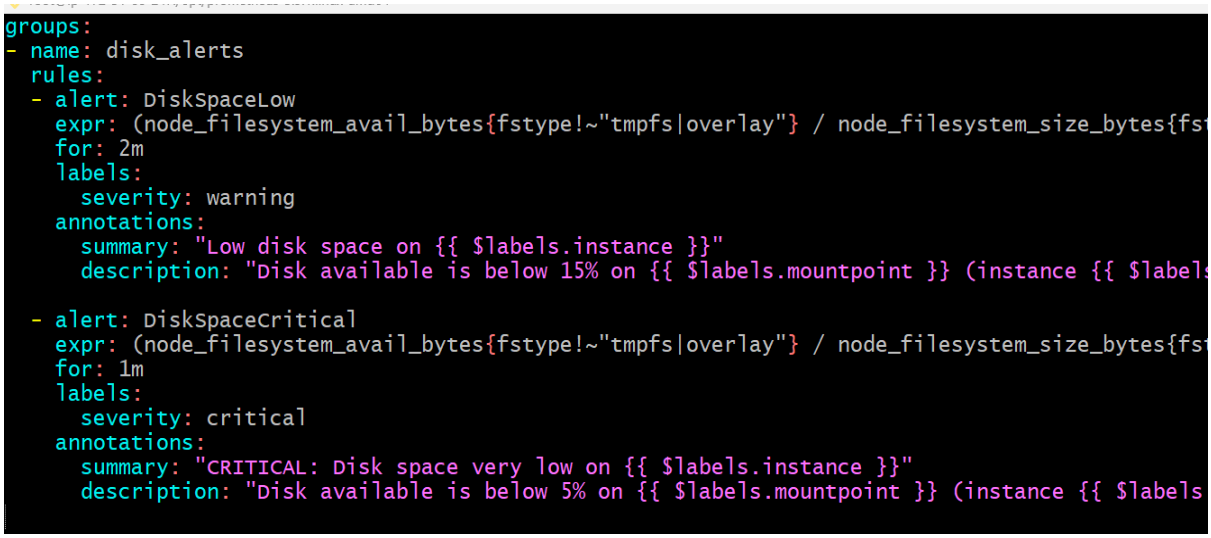
labels:

severity: critical

annotations:

summary: "CRITICAL: Disk space very low on {{ $labels.instance }}"

description: "Disk available is below 5% on {{ $labels.mountpoint }} (instance {{ $labels.instance }})"



```
groups:
- name: disk_alerts
  rules:
  - alert: DiskSpaceLow
    expr: (node_filesystem_avail_bytes{fstype!~"tmpfs|overlay"} / node_filesystem_size_bytes{fs
    for: 2m
    labels:
      severity: warning
    annotations:
      summary: "Low disk space on {{ $labels.instance }}"
      description: "Disk available is below 15% on {{ $labels.mountpoint }} (instance {{ $label

  - alert: DiskSpaceCritical
    expr: (node_filesystem_avail_bytes{fstype!~"tmpfs|overlay"} / node_filesystem_size_bytes{fs
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "CRITICAL: Disk space very low on {{ $labels.instance }}"
      description: "Disk available is below 5% on {{ $labels.mountpoint }} (instance {{ $labels
```

Link this file to Prometheus.yml file

- - “disk_alerts.yml”

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is eve
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
           - "localhost:9093"

# Load rules once and periodically evaluate them according to the global 'evaluation
rule_files:
  - "rules/*.yml"
  - "disk_alerts.yml"

  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
      - targets: ["localhost:9100"]
      - targets: ["localhost:9116"]
       # The label name is added as a label `label_name=<label_value>` to any timese
```

- **/opt/prometheus-3.9.1.linux-amd64/promtool check config /opt/prometheus-3.9.1.linux-amd64/prometheus.yml**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# /opt/prometheus-3.9.1.linux-am
ux-amd64/prometheus.yml
Checking /opt/prometheus-3.9.1.linux-amd64/prometheus.yml
  SUCCESS: 2 rule files found
 SUCCESS: /opt/prometheus-3.9.1.linux-amd64/prometheus.yml is valid prometheus confi

Checking /opt/prometheus-3.9.1.linux-amd64/rules/alerts.yml
  SUCCESS: 2 rules found

Checking /opt/prometheus-3.9.1.linux-amd64/disk_alerts.yml
  SUCCESS: 2 rules found

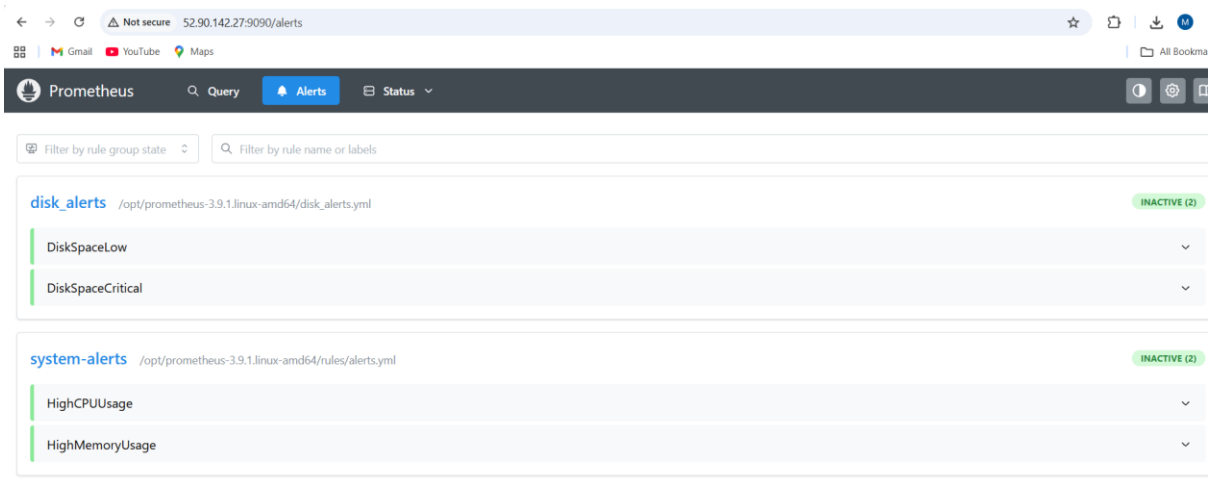
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]#
```

- **systemctl restart prometheus**
- **systemctl status prometheus**

```
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl restart prometheus
[root@ip-172-31-88-247 prometheus-3.9.1.linux-amd64]# systemctl status prometheus
● prometheus.service - Prometheus Monitoring
     Loaded: loaded (/etc/systemd/system/prometheus.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-01-21 14:33:54 UTC; 7s ago
   Main PID: 11198 (prometheus)
      Tasks: 8 (limit: 1067)
     Memory: 34.7M
        CPU: 222ms
     CGroup: /system.slice/prometheus.service
             └─11198 /opt/prometheus-3.9.1.linux-amd64/prometheus --config.file=/opt/prometheus

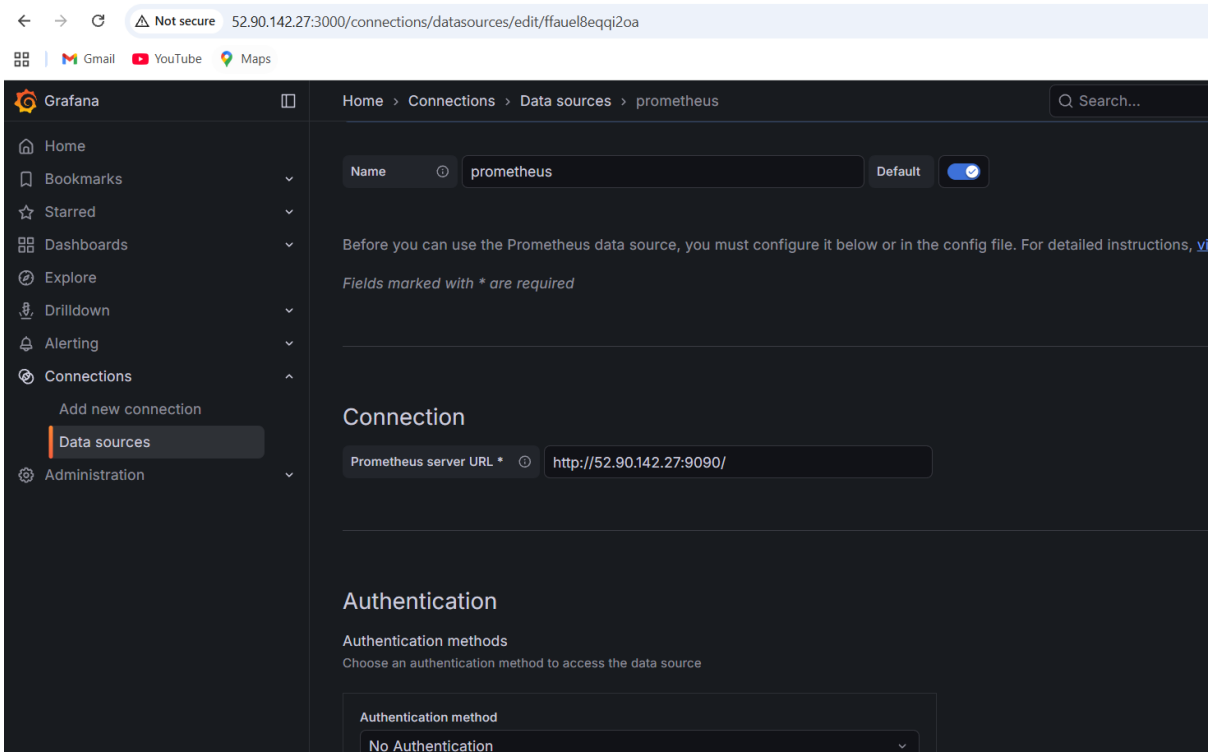
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.218Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.253Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.253Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.253Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.258Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.258Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.258Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.263Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.263Z
Jan 21 14:33:54 ip-172-31-88-247.ec2.internal prometheus[11198]: time=2026-01-21T14:33:54.263Z
lines 1-20/20 (END)
```

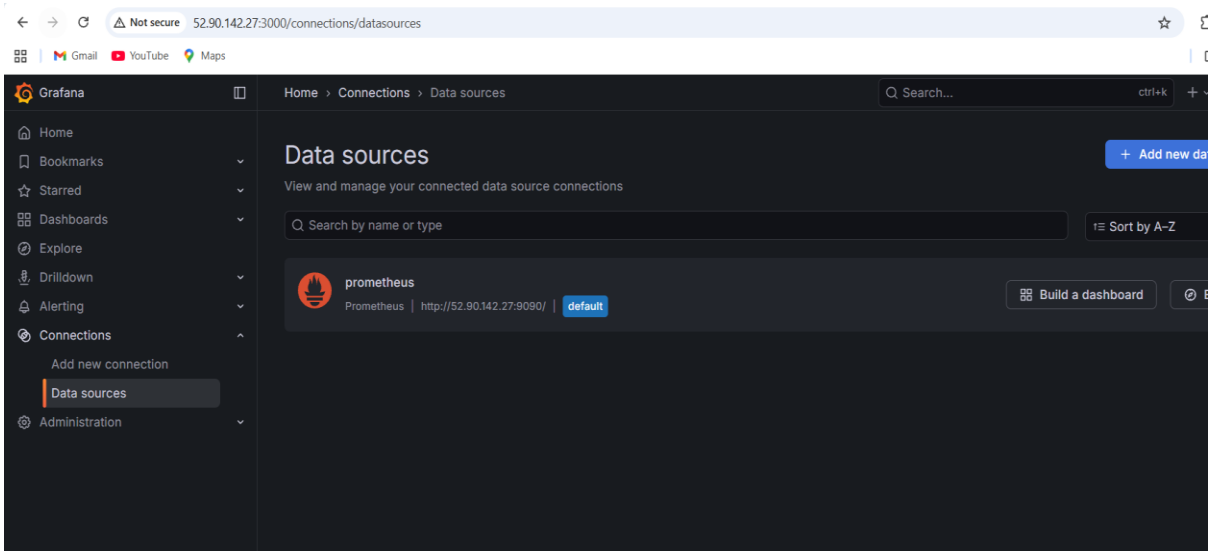
Search in browser with the ip and port 9090:alerts



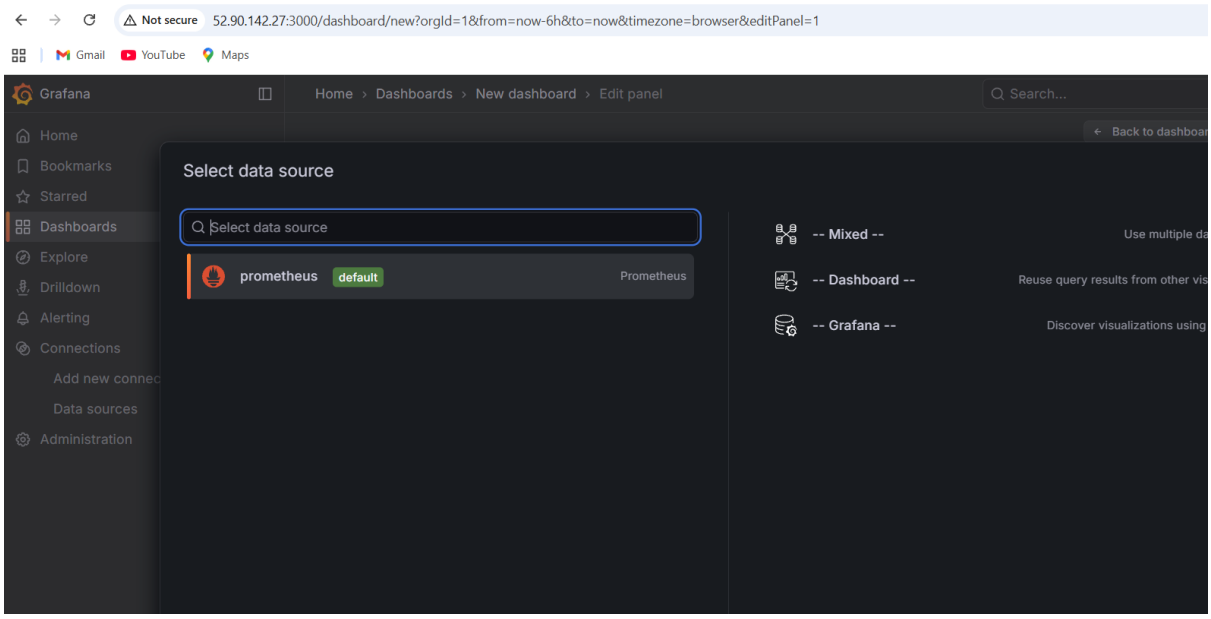
9. Create a custom Grafana dashboard using Prometheus metrics for specific insights.

Go to data source select Prometheus as data source



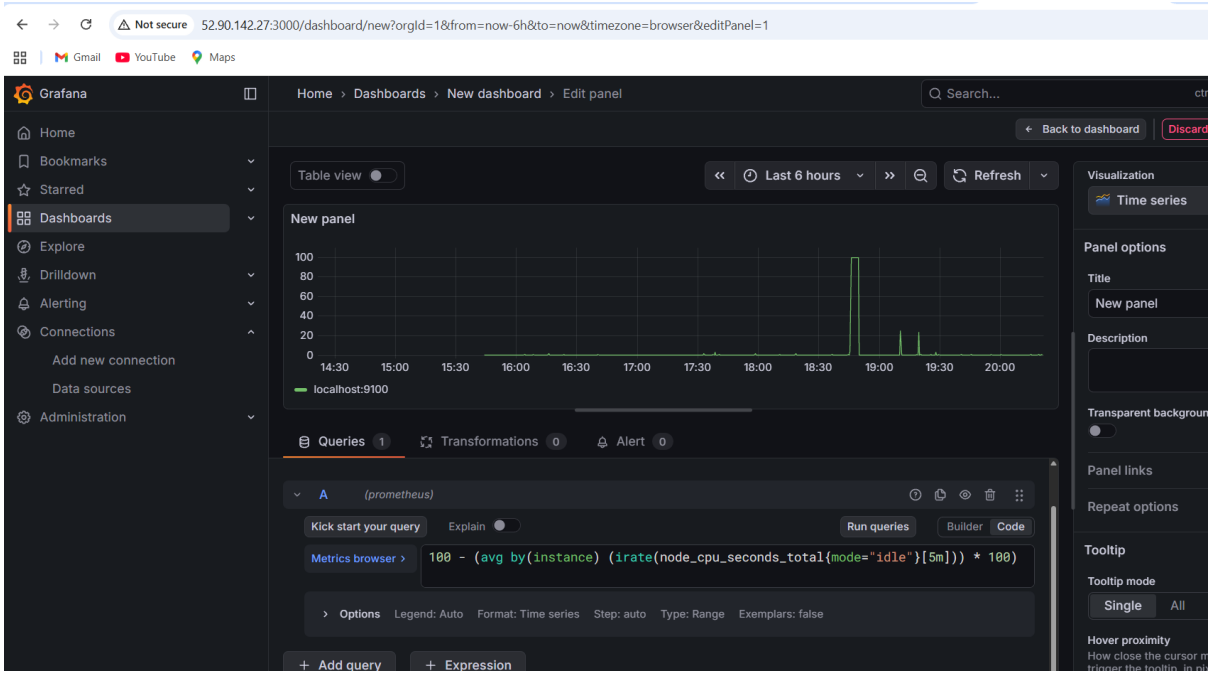


Click on new dashboard and select Prometheus



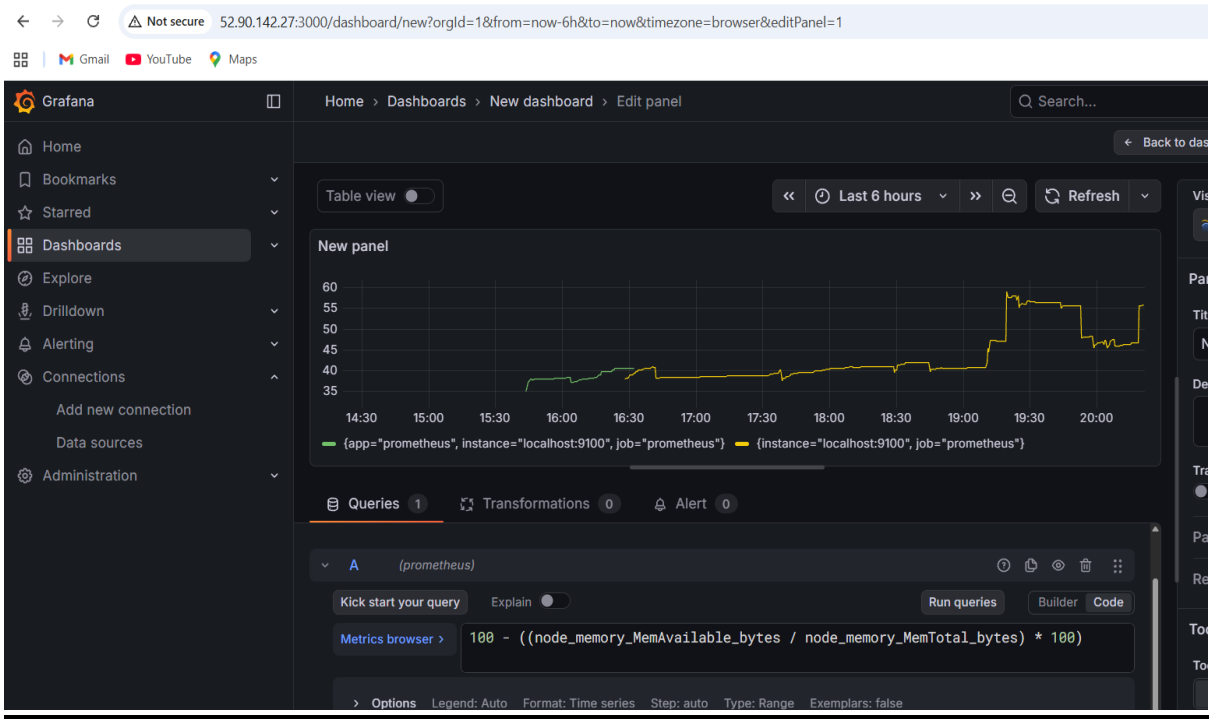
Cpu:

- **100 - (avg by(instance) (irate(node_cpu_seconds_total{mode="idle"}[5m])) * 100)**



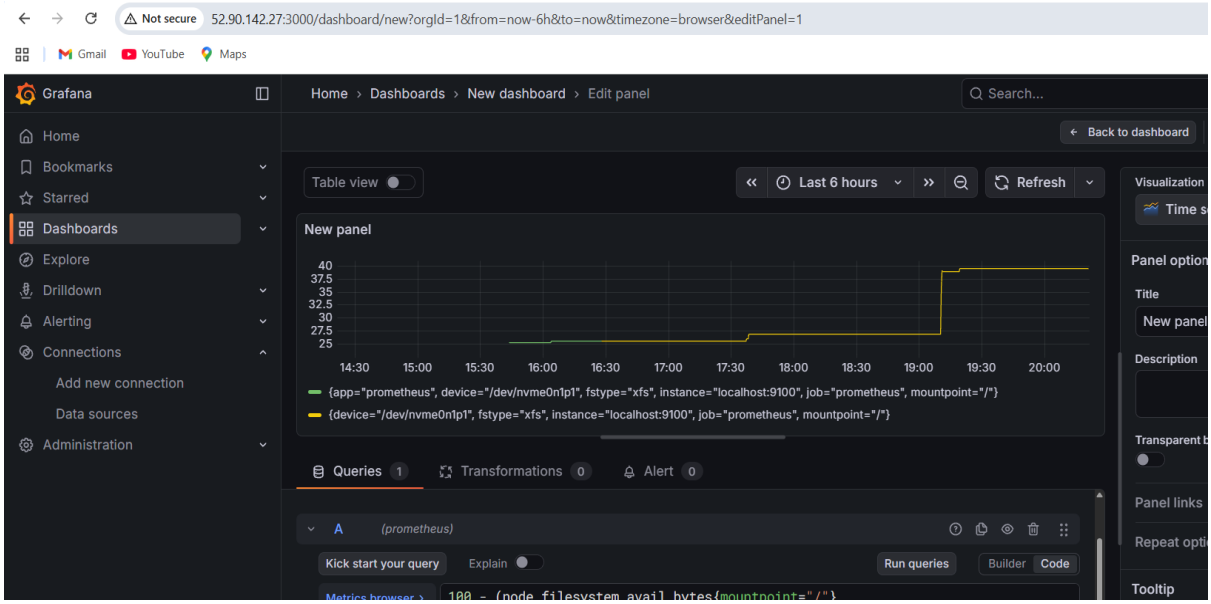
Memory:

- **100 - ((node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes) * 100)**



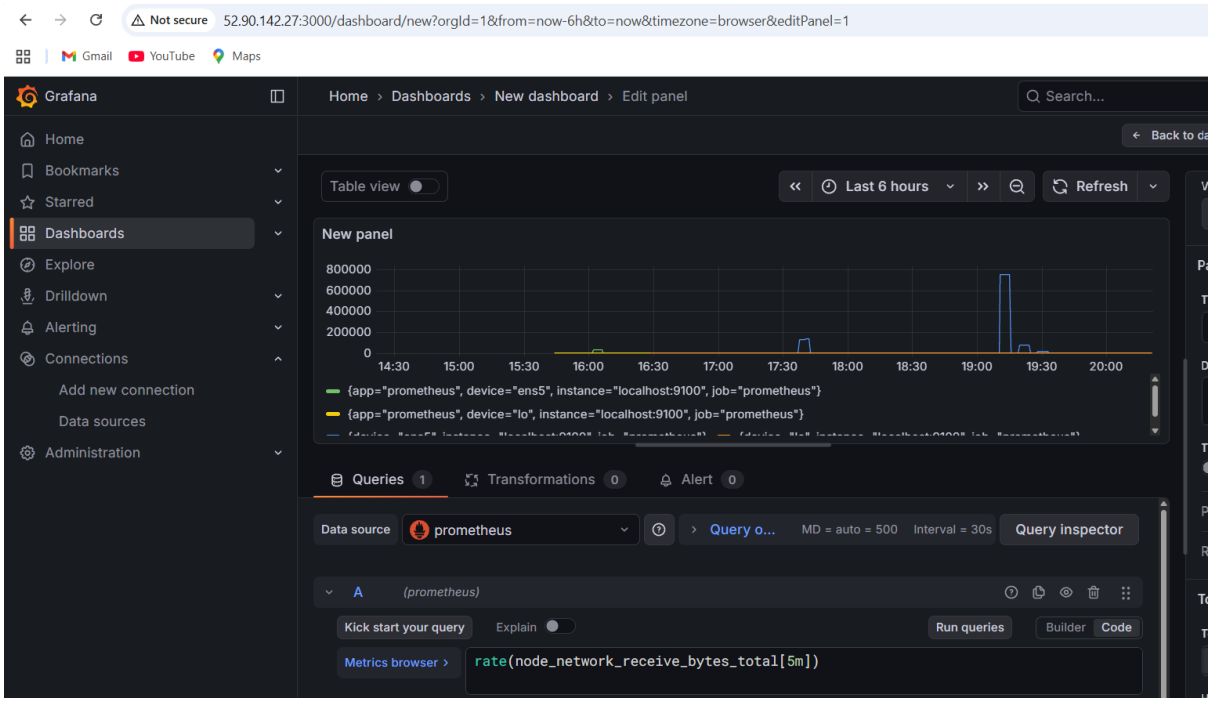
Disk usage:

- **100 - (node_filesystem_avail_bytes{mountpoint="/"} / node_filesystem_size_bytes{mountpoint="/"} * 100)**



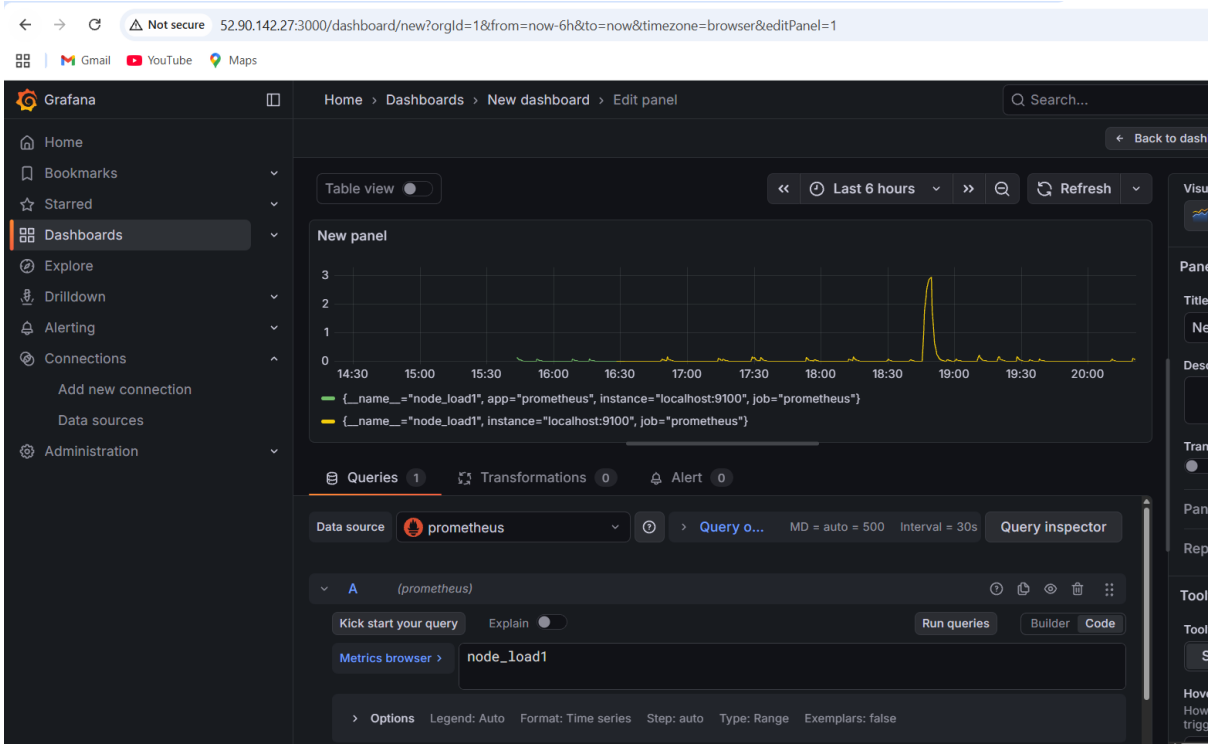
Network traffic:

- **rate(node_network_receive_bytes_total[5m])**

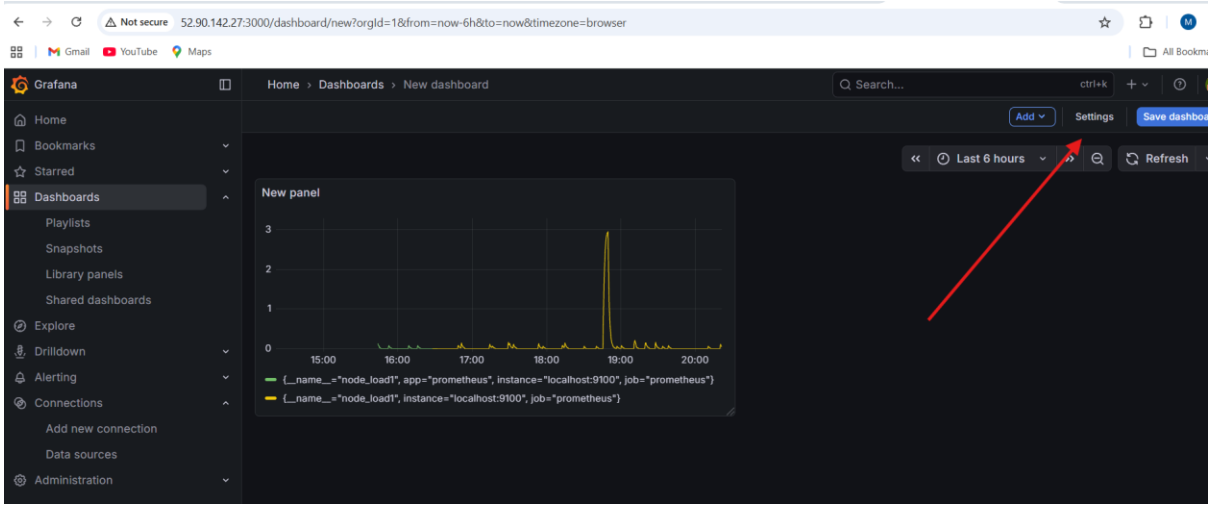


System load:

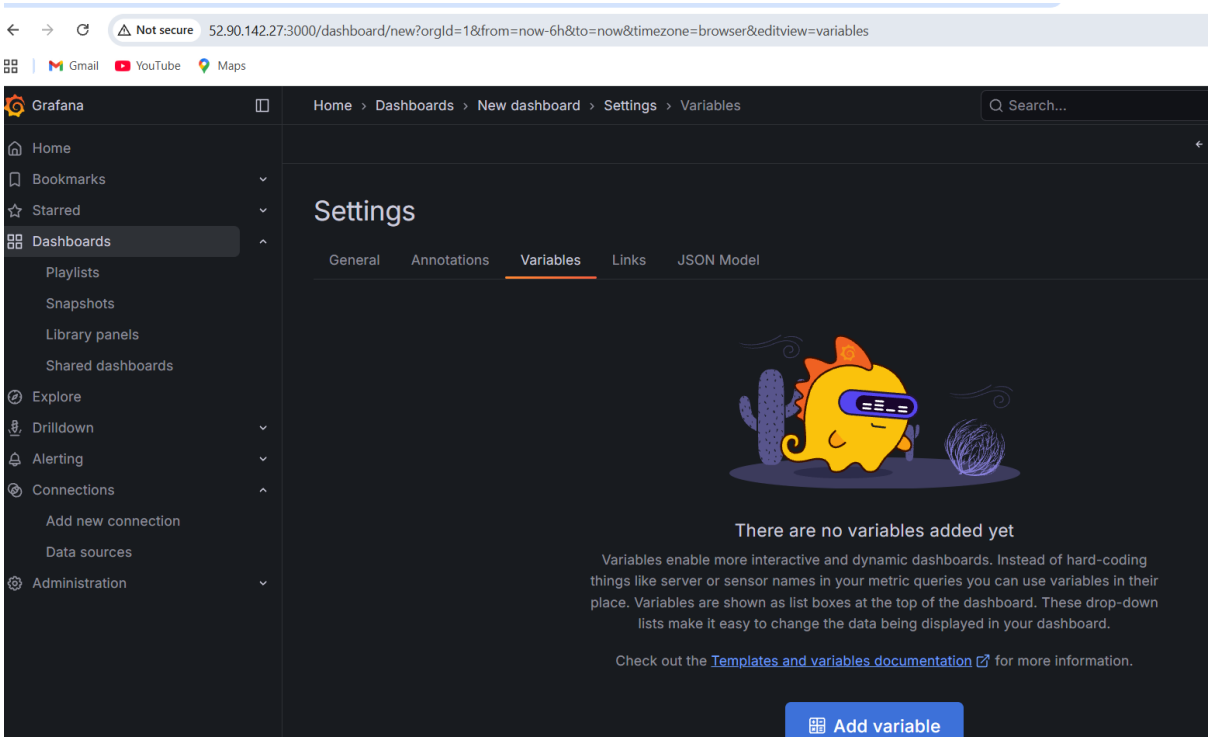
- **node_load1**



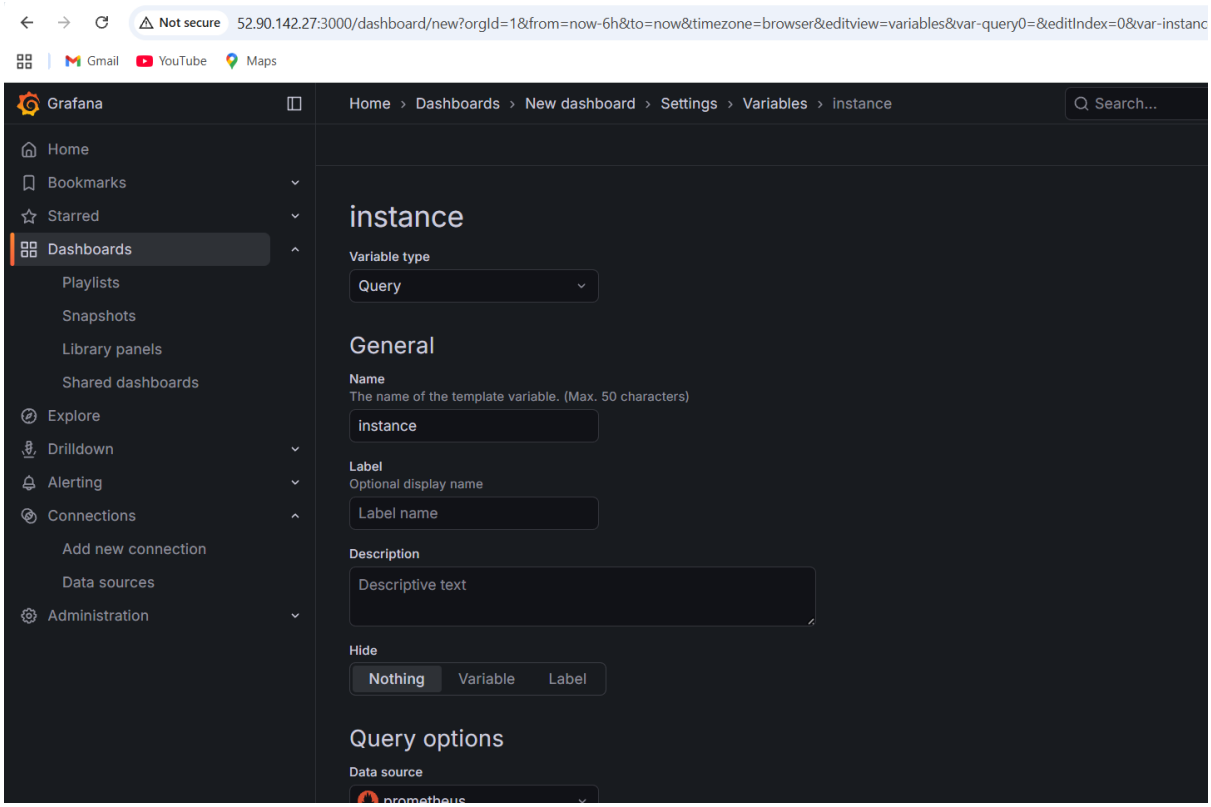
Click on dashboardsettings



Click on variable add variable



Add variable name as instance and give a query anything you want and click on save dashboard then you will see your dashboard.


Not secure
52.90.142.27:3000/dashboard/new?orgId=1&from=now-6h&to=now&timezone=browser&editview=variables&var-query0=&editIndex=0&var-instanc
Gmail
YouTube
Maps
Grafana
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Home › Dashboards › New dashboard › Settings › Variables › instance
Search...
instance
Variable type
Query
General
Name
The name of the template variable. (Max. 50 characters)
instance
Label
Optional display name
Label name
Description
Descriptive text
Hide
Nothing
Variable
Label
Query options
Data source
prometheus